

Polyconvex neural-network surrogates with learned paired directional features for anisotropic hyperelastic homogenization

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Abstract

Approximating the effective anisotropic hyperelastic response of a representative volume element (RVE) requires constitutive expressiveness, mechanical consistency, and computational efficiency. Non-intrusive neural surrogates provide direct constitutive evaluations, whereas intrusive projection-based reduced-order models (PROMs) approximate the microscopic state and equilibrium equations. This work develops a polyconvex physics-augmented neural network (PANN) construction based on paired directional features with trainable orientations, stretch powers, and determinant exponents. The features are convex in independent deformation minors and are composed with convex, componentwise nondecreasing cores realized as an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN). The complete energies satisfy objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and prescribed growth conditions by construction; global nonnegative energy is assessed through a separate sufficient bound for the trained parameters. On a periodic cell with four nonaligned cavities, learning six paired features lowers the held-out stress error sevenfold, to about 0.04%. On a second cell, with a rotated elliptical void, the selected polyconvex models attain approximately 0.63%. An unconstrained energy trained identically is more accurate but, beyond the data, loses rank-one convexity and remains finite under volume collapse. In a tensile-coupon FE^2 application against a full-order reference, the ICNN and ICKAN give displacement errors below 0.42%, and hyperreduced PROMs provide a complementary accuracy–cost benchmark. Under confined compression beyond the data, the polyconvex energies reach 500 MPa, whereas the unconstrained energy reaches less than half that load, where its rank-one curvature turns negative. These results quantify what the guarantees cost in accuracy and what they secure beyond the data.

Keywords: polyconvexity; physics-augmented neural networks; anisotropic hyperelasticity; computational homogenization; FE^2 ; input-convex neural networks; projection-based reduced-order models.

1 Introduction

The effective response of a heterogeneous material emerges from the deformation of its constituents and their interactions with the microstructure. At finite strains, this interaction can produce anisotropy and coupling between loading modes that are difficult to capture with a predetermined macroscopic constitutive law. Computational homogenization addresses this difficulty by resolving a representative volume element (RVE), in which an imposed deformation drives a microscopic boundary-value problem whose solution yields the effective stresses and constitutive derivatives [1–3]. The effective law is thus computed rather than prescribed in advance.

This flexibility comes at a computational cost. Characterizing the material over a range of deformations already requires many nonlinear finite element solves; using the same RVE within a structural calculation multiplies those solves across macroscopic integration points and load increments [3, 4]. Much of this repeated work concerns closely related microscopic states. Surrogate modeling seeks to exploit that relationship by computing representative solutions offline and reusing their information in less expensive online evaluations. The goal is to reduce the cost of these evaluations without losing the constitutive behavior that made the RVE description useful.

Two complementary approximation routes are relevant, but they play different roles in this work. Non-intrusive constitutive surrogates learn the effective response from RVE input–output data and dispense with

the microscopic solve online [5]. Intrusive projection-based reduced-order models (PROMs) instead compress the microscopic displacement field and retain projected equilibrium equations [6, 19, 20], with hyperreduction limiting the local contributions required to evaluate them [34]. This work follows the first route and learns the effective stored energy while embedding the constitutive restrictions needed for its mechanical use. The intrusive route is retained as a secondary reference because it preserves an approximate microscopic state and the underlying constituent law.

The central contribution is a family of learnable paired directional features for polyconvex anisotropic energies. Their orientations and exponents adapt to the RVE response, while their paired construction makes the stress vanish exactly at the undeformed state without compromising polyconvexity. Coupling these features to either an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN) gives two realizations of the same constitutive design. PROM-based surrogates of the same RVE then provide secondary accuracy–cost references. Two structural examples follow, a tensile coupon and a confined compression test beyond the data, to show how the learned energies behave once they leave isolated material evaluations.

1.1 Energy-based constitutive learning and polyconvexity

Direct constitutive learning moves the expensive microscopic solves to the data-generation stage and replaces them online by a trained material model. Neural homogenization has demonstrated this separation [7], making systematic sampling of representative microscopic responses an essential part of the offline workflow [8]. What the surrogate must represent depends on the material class. History-dependent responses may require learned history-to-response maps [9] or internal descriptors such as maximum directional stretches [10], while interacting scales may motivate staggered data generation and training [11]. The present work, by contrast, considers history-independent hyperelasticity. This removes memory from the state but requires the stress to derive from a stored-energy function.

Learning that function, rather than fitting each stress component independently, makes the stress work conservative and gives a major-symmetric constitutive tangent by differentiation. This is a central motivation for the mechanics-informed energy formulation of As’ad et al. [12]. Expressing the energy through objective deformation measures then removes dependence on a superposed rigid rotation. In this way, physics-augmented neural networks (PANNs) shift part of the burden from learning mechanical identities from data to satisfying them through the representation itself [13, 14]. Prediction accuracy remains to be established, but potential structure and objectivity need not depend on how densely the training domain has been sampled.

Preserving a potential does not, by itself, control its curvature. At finite strains, that control must also be imposed in appropriate variables. Convexity in Green–Lagrange strain [12] is attractive because stress and tangent follow directly from the learned energy. Its implication for curvature with respect to the deformation gradient nevertheless contains a stress-dependent term [15], so it is not a global polyconvexity guarantee. Conversely, Xu et al. [16] parameterize an incremental tangent through a lower-triangular factorization to enforce positive semidefiniteness. This controls the tangent matrix but does not define the scalar potential required for integrability. The modeling task is therefore to preserve both the potential and the desired finite-strain curvature condition within one construction.

Polyconvexity provides a route to that combination. It represents the energy as a convex function of deformation minors, yielding a sufficient condition for rank-one convexity [17]. For the two-dimensional model considered here, the independent arguments are the deformation gradient and its determinant. This framework also underpins existence results when the additional coercivity and admissibility hypotheses are met [18]. Polyconvexity serves here as a sufficient constitutive restriction, not as a necessary property of every homogenized material or a guarantee of structural uniqueness. The remaining challenge is to keep enough freedom within these restrictions to describe the anisotropic RVE response.

1.2 Directional features and representational expressiveness

For anisotropy, this challenge is especially consequential, because a flexible scalar core can distinguish only the deformation states that its inputs distinguish. Structural tensors encode material directions in objective scalar quantities and underpin established anisotropic polyconvex energies [40]. The directional description is itself a constitutive choice. Models of fiber-reinforced tissue, for example, must account for a distribution of orientations rather than one perfectly aligned family when that dispersion affects the mechanical response [41]. A learned feature map must likewise expose the relevant directional distinctions while remaining compatible with objectivity and convexity in the deformation minors. It is therefore part of the constitutive model, not merely a preprocessing step.

A convex scalar core does not resolve this choice. An ICNN provides a convex function [42], but a sufficient composition with nonlinear features requires those features to be convex in the independent deformation

minors and the core to be componentwise nondecreasing [13]. This monotonicity imposes an order on feature space, since $\mathbf{z}(\mathbf{F}_a) \leq \mathbf{z}(\mathbf{F}_b)$ componentwise implies $H(\mathbf{z}(\mathbf{F}_a)) \leq H(\mathbf{z}(\mathbf{F}_b))$. Increasing the network width cannot reverse that ordering. Reference corrections and growth terms, added later to complete the energy, do not remove this restriction on the core contribution. A poorly chosen feature family can therefore impose a representational obstruction before network capacity or optimization becomes limiting.

The numerical evidence of Klein et al. [13] illustrates this distinction. For their cubic beam-lattice data, an exactly objective invariant-based polyconvex model fails to represent the shear response, whereas a model formulated in $\mathbf{F}$, $\text{cof } \mathbf{F}$, and $\det \mathbf{F}$ gives a substantially better fit while approximating objectivity through data augmentation. The same invariant-based model performs very well for their analytically generated transversely isotropic data. In these results, accuracy depends on whether the admissible inputs expose the distinctions required by the material response, rather than on polyconvexity alone. This motivates enlarging the feature family before concluding that a convex core is inadequate.

The natural response is to make the directional description adaptive without abandoning the admissible composition. Generalized-structure-tensor models calibrate the structural tensors together with the energy model, allowing the data to identify aspects of the anisotropy [43]. Recent triclinic invariant constructions and group symmetrization further broaden the polyconvex descriptions available for finite material-symmetry groups [44]. Greater directional freedom, however, must be reconciled with the undeformed state. A learned anisotropic contribution need not produce zero reference stress, while cancelling that stress with an arbitrary strain correction can violate polyconvexity or material symmetry. The invariant-based normalization and analytic growth terms of Linden et al. [14] demonstrate why these operations must belong to the admissible constitutive construction. The feature family must therefore do two jobs at once: distinguish the relevant directional response and control its reference derivatives so that exact stress normalization remains compatible with polyconvexity. This is the role of the paired features introduced here.

While the feature map determines what the model can distinguish, the scalar core determines how those distinctions are combined into an energy. Prior constructions distribute this flexibility in different ways. Constitutive artificial neural networks (CANNs) learn coefficients and exponents within an interpretable catalog of admissible energy terms [45], whereas neural ordinary differential equations can learn monotone energy derivatives and recover convex energy contributions by integration [46]. In the present work, an ICNN combines the paired features through nonnegative weights and fixed convex activations. A Kolmogorov–Arnold network (KAN) instead places trainable univariate functions on its edges, a flexibility explored in constitutive KANs [47]. The feature-composition argument holds only if these edge functions are convex and nondecreasing, as in monotone input-convex KANs (ICKANs) [48]. A smooth ICNN and an integrated-hat ICKAN are compared under the same paired features and analytic completion. The comparison isolates how the scalar dependence is represented, while the directional information and the mechanical guarantees stay the same.

1.3 Projection-based reduced-order models

The preceding approach learns the effective energy and thereby removes the microscopic solve from the online stage. Projection-based reduction makes a different compromise, retaining the RVE boundary-value problem but seeking its solution in a space learned from previous microscopic states. In an affine PROM, those states define a basis, commonly obtained through proper orthogonal decomposition (POD), and projected equilibrium determines the coordinates of each new displacement field. Reduced finite-strain homogenization has shown that this strategy can reuse information from earlier RVE solves without replacing the constituent law [19–21]. PROMs therefore provide a natural physics-based reference for the present study, since they preserve more of the microscopic problem than an effective-energy surrogate at the price of an online equilibrium solve.

The affordability of that solve depends first on the geometry of the solution family. A small number of loading parameters does not necessarily imply that the corresponding microscopic states lie close to a small global linear space. When their Kolmogorov n -width decays slowly, many basis vectors, and hence many equilibrium unknowns, are required to achieve adequate accuracy. Considerable effort has been devoted to mitigating this Kolmogorov barrier with richer reduced representations, from local linear spaces [25] to nonlinear manifolds built with autoencoders [26], quadratic corrections [27], or learned closures in a POD coordinate space [22–24, 50]. The last of these, PROM-ANN [22], offers the compromise sought here, because it enriches the geometry of the reduced state without giving up either its POD description or projected equilibrium. Equilibrium determines only a small set of primary coordinates, while a regression model, here an artificial neural network (ANN), reconstructs the secondary coordinates that populate a much richer POD expansion [24]. The microscopic displacement can thereby draw on many modes while the online nonlinear system remains small, which motivates the primary–secondary construction adopted for the RVE in this work.

Reducing the number of unknowns does not make the reduced equations inexpensive, because assembling their residual and tangent may still require constitutive evaluations over the complete microscopic mesh at every iteration. While hardware acceleration performs this full-mesh work faster [28], hyperreduction reduces how much of it must be performed, turning a PROM into a hyperreduced PROM (HPROM). Methods that act directly on the projected equations [29, 30], instead of first reconstructing high-dimensional quantities as discrete empirical interpolation does [31], include the linear-programming empirical quadrature procedure [35, 36], energy-conserving sampling and weighting (ECSW) [32, 33], and the empirical cubature method (ECM) [21, 34]. Each replaces full assembly by a nonnegative weighted sum of selected element contributions while leaving the local constituent evaluations intact. This work follows ECM because previous developments extend its fixed rule, in which one support and one weight vector serve the whole training space, toward movable points [37], subspace-adaptive weights [38], and finally the manifold-adaptive weights ECM (MAW-ECM) [39], which is the path needed to couple hyperreduction with PROM-ANN. In MAW-ECM, the weights become fields over the latent coordinates, so the same primary coordinates that locate the microscopic state on the PROM-ANN manifold also select the rule used to evaluate its equilibrium and stress.

Together, these choices yield three intrusive references for the structural comparison. The affine HPROM retains a linear POD state, solves projected equilibrium, and evaluates it with fixed ECM. HPROM-ANN replaces the linear state by the primary-secondary manifold and the fixed rule by MAW-ECM, but still determines its primary coordinates through equilibrium. Its direct variant, D-HPROM-ANN, removes that final online solve by evaluating the decoder at the imposed strain, as in the hyperelastic example of [39]. Moving across the three models removes successive portions of the online microscopic work while preserving an approximate microscopic state and selected evaluations of the constituent law. They contextualize the accuracy and cost of the proposed energy surrogates but do not share their constitutive guarantees, because the approximate effective stress of a reduced model need not derive from a single effective potential, even when the constituent law is retained and the cubature weights are nonnegative.

1.4 Present contribution

Against these intrusive physics-based references, the central route pursued in this work is an effective-energy surrogate whose anisotropy is learned without relinquishing polyconvexity or exact reference-state normalization. The construction rests on paired directional features with trainable orientations, stretch powers, and determinant exponents constrained by explicit joint-convexity bounds.

The pairing is designed so that the strain derivative of every feature at the undeformed state is proportional to the identity tensor, even though its response away from that state remains anisotropic. The complete learned contribution can consequently be normalized by a single scalar correction affine in the determinant. This correction cancels the reference stress without tying that cancellation to the independently adjustable volumetric stiffness, allowing the directional description to adapt to the RVE while the reference state remains under analytic control.

With the directional information in this form, two monotone convex cores provide alternative representations of its scalar energetic dependence: a smooth ICNN and an integrated-hat spline ICKAN. In both cases, the learned core, reference correction, and growth terms form one complete energy for which the analysis establishes objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and the specified growth limits. Global nonnegative energy is treated separately through a sufficient lower bound evaluated for the trained parameters. The novelty lies in bringing established convex-network and affine-determinant normalization principles together through the paired anisotropic family and its explicit analytical conditions.

The numerical study then compares these models with the Unconstrained energy, a network trained under the same protocol without the curvature and growth constraints, to interpret the cost of the guarantees, and with the reduced microscopic evaluators just described, to contextualize structural accuracy and online cost.

1.5 Assessment strategy and scope

The assessment follows the two levels at which the surrogates are intended to operate. At the constitutive level, it asks how accurately the constrained energies represent independent anisotropic RVE responses and what is gained by learning the paired directions rather than prescribing them. Response errors and the fixed/learned-feature comparison address approximation quality, while derivative checks and saved-parameter energy bounds interrogate behavior that a fit metric alone cannot reveal [49]. Sampled curvature tests check the implementation of the constrained energies, whose polyconvexity follows from the analytical construction, and locate where the Unconstrained energy loses rank-one convexity beyond the data.

At the structural level, the question becomes whether those constitutive approximations remain accurate and useful after being embedded in a nonlinear boundary-value problem. A tensile coupon provides this deployment test, and a confined compression test then drives the same energies beyond the data to ask which

of them keep the structural problem solvable. Within the first-order two-level finite element formulation, denoted FE^2 , an RVE problem supplies the material response at each macroscopic integration point [4]. Here, every point queries the same periodic cell. Replacing that query by a surrogate propagates its approximation through a common equilibrium problem. A nested full-order model (FOM)– FE^2 calculation supplies the reference, while field-level comparisons complement aggregate displacement and stress errors. The intrusive reduced models then provide physics-based accuracy–cost benchmarks without turning the development of a new FE^2 scheme into a second objective of the paper.

These two levels deliberately separate analytical guarantees from empirical adequacy. The constitutive construction establishes the stated properties of the complete energy, whereas independent RVE and structural evaluations assess how well that representation approximates the selected material and how much online work it avoids. Additional closed-cycle and curvature diagnostics are documented in the supplementary material.

The resulting claims are tied to the scope of the study: the two periodic cells, plane-strain kinematics, and the two structural loadings. Agreement with FOM– FE^2 measures surrogate error within the adopted homogenization model, not error relative to a structure in which every microscopic cell is resolved explicitly. Reported timings cover the online workflows, whereas offline data generation and training are not included. Within these boundaries, the objective is to establish the usefulness and constitutive properties of the proposed representation.

With this assessment strategy established, Section 2 defines the homogenization problem and work-conjugate variables. Section 3 introduces the admissibility criteria, constructs the learned energies, and proves their properties, after which Section 4 defines the reduced microscopic references used in the structural comparison. Section 5 reports the numerical examples, first the multicavity RVE (MC-RVE) representation study and then the single-cavity RVE (SC-RVE) deployment study, which ends with the coupon FE^2 comparison and a confined compression test beyond the data. Section 6 closes the paper.

2 Finite-strain homogenization and the reference problem

The two surrogate routes replace different parts of the online calculation, but both approximate the same constitutive map: for a prescribed macroscopic deformation, determine the effective response of a periodic RVE. The unreduced RVE model supplies the reference by solving microscopic equilibrium. This section establishes the kinematic input, periodic cell problem, effective outputs, and stress–tangent interface shared by all models before their approximation strategies diverge. The cell geometry and material parameters are specified in the numerical examples.

2.1 Kinematics and work-conjugate variables

Let $\mathbf{F} \in \text{GL}^+(2)$ denote the prescribed in-plane macroscopic deformation gradient. Its associated right Cauchy–Green tensor and Green–Lagrange strain are

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad J = \det \mathbf{F} > 0. \quad (1)$$

Here $\mathbf{C}$ and $\mathbf{E}$ describe the macroscopic state. The macroscopic input is written simply as $\mathbf{F}$ to match the constitutive notation used throughout the paper. The subscript μ marks microscopic quantities, whereas the unsubscripted W , $\mathbf{P}$, and $\mathbf{S}$ denote effective quantities obtained by homogenizing the cell. We adopt plane strain: the corresponding three-dimensional gradient is $\text{diag}(\mathbf{F}, 1)$, and the same embedding applies locally to $\mathbf{F}_\mu$. This constrains the out-of-plane deformation, not the out-of-plane stress.

The constitutive input used throughout is the three-component engineering representation of the Green–Lagrange strain. With $\mathbf{S}$ the effective second Piola stress, the work-conjugate convention is

$$\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T, \quad \mathbf{s} = (S_{11}, S_{22}, S_{12})^T, \quad \mathbf{S} : d\mathbf{E} = \mathbf{s} \cdot d\mathbf{e}. \quad (2)$$

Thus $\gamma_{12} = 2E_{12}$ is paired with S_{12} , with no additional factor of two in the stress vector. For a differentiable effective energy density $W(\mathbf{e})$, this convention gives the stress and the constitutive tangent

$$\mathbf{s} = \frac{\partial W}{\partial \mathbf{e}}, \quad \mathbf{D} = \frac{\partial \mathbf{s}}{\partial \mathbf{e}} = \frac{\partial^2 W}{\partial \mathbf{e} \partial \mathbf{e}} \in \mathbb{R}^{3 \times 3}, \quad D_{ij} = \frac{\partial s_i}{\partial e_j}. \quad (3)$$

This convention is used consistently in training, work integrals, and tangent checks.

An admissible strain input must satisfy $\mathbf{I} + 2\mathbf{E} \succ 0$. We query the RVE with the representative

$$\mathbf{F}(\mathbf{e}) = \sqrt{\mathbf{I} + 2\mathbf{E}}, \quad (4)$$

where the square root is symmetric positive definite. Objectivity allows a superposed macroscopic rotation to be removed without changing the energy or second Piola stress. This choice does not assume isotropy: material directions remain fixed in the reference configuration. Nor does it require the local microscopic gradients to be symmetric.

2.2 Periodic microscopic equilibrium

Let Y denote the reference periodic cell and $Y_s \subset Y$ its solid part. The prescribed $\mathbf{F}$ determines the affine part of the microscopic deformation; the fluctuation $\mathbf{w}$ is unknown:

$$\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{F}\mathbf{X} + \mathbf{w}(\mathbf{X}), \quad \mathbf{F}_\mu = \mathbf{F} + \nabla \mathbf{w}, \quad (5)$$

with gradients taken in the reference configuration. Periodicity requires $\mathbf{w}(\mathbf{X}^+) = \mathbf{w}(\mathbf{X}^-)$ at paired points on opposite cell faces. Consequently, the microscopic displacement $\mathbf{u}_\mu = \boldsymbol{\varphi} - \mathbf{X}$ satisfies

$$\mathbf{u}_\mu(\mathbf{X}^+) - \mathbf{u}_\mu(\mathbf{X}^-) = (\mathbf{F} - \mathbf{I})(\mathbf{X}^+ - \mathbf{X}^-). \quad (6)$$

The finite-strain displacement jump therefore depends on $\mathbf{F} - \mathbf{I}$, not directly on $\mathbf{E}$. Setting one fluctuation value to zero removes the arbitrary translation; it does not clamp the entire cell boundary. The internal void surface is traction-free.

Let $\psi_\mu(\mathbf{X}, \mathbf{F}_\mu)$ be the solid's objective hyperelastic energy per unit reference volume and $\mathbf{P}_\mu = \partial_{\mathbf{F}_\mu} \psi_\mu$ its first Piola stress. In the absence of microscopic body forces, equilibrium requires

$$\int_{Y_s} \mathbf{P}_\mu : \nabla \delta \mathbf{w} \, dV = 0 \quad (7)$$

for every admissible periodic test fluctuation. Periodic kinematics and the associated traction balance provide the standard work-consistent cell problem [1]. After finite element discretization and elimination of the periodic constraints, let $\mathbf{a}$ collect the independent fluctuation coordinates. The full-order equilibrium problem is

$$\mathbf{r}(\mathbf{a}, \mathbf{e}) = \partial_{\mathbf{a}} \Pi_\mu(\mathbf{a}, \mathbf{e}) = \mathbf{0}, \quad \Pi_\mu = \int_{Y_s} \psi_\mu(\mathbf{X}, \mathbf{F}_\mu) \, dV, \quad (8)$$

where the strain dependence enters through Eq. (4). Solving Eq. (8) gives $\mathbf{a}^*(\mathbf{e})$ and hence the microscopic fields. The microscopic deformation must also remain orientation-preserving, $\det \mathbf{F}_\mu > 0$.

2.3 Effective energy and stress

On the selected equilibrium branch, the effective energy and stresses are

$$W(\mathbf{e}) = \frac{\Pi_\mu(\mathbf{a}^*(\mathbf{e}), \mathbf{e})}{|Y|}, \quad \mathbf{P} = \frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu \, dV, \quad \mathbf{S} = \mathbf{F}^{-1} \mathbf{P}. \quad (9)$$

The normalizing volume $|Y|$ includes the void: these are properties of the porous cell, not of its solid fraction alone. For the two-dimensional discretization, the same reference thickness enters the integration weights and the cell volume, and cancels in their ratio. The stress conversion in Eq. (9) is performed *after* averaging the first Piola stress. In general, averaging microscopic second Piola stresses does not produce $\mathbf{S}$, because $\mathbf{F}_\mu$ varies within the cell.

The connection between energy and averaged stress follows from equilibrium. For a macroscopic variation, $\delta \mathbf{F}_\mu = \delta \mathbf{F} + \nabla \delta \mathbf{w}$; Eq. (7) removes the fluctuation contribution and yields

$$\frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu : \delta \mathbf{F}_\mu \, dV = \mathbf{P} : \delta \mathbf{F} = \mathbf{S} : \delta \mathbf{E} = \mathbf{s} \cdot \delta \mathbf{e}. \quad (10)$$

This is the Hill–Mandel work-consistency condition [1]; it also establishes $\mathbf{s} = \partial_{\mathbf{e}} W$ on a smooth branch.

The Newton solver follows an equilibrium branch, possibly through load continuation. Convergence establishes stationarity, not global minimization of Π_μ or uniqueness of the cell response. We use the resulting smooth branch as the reference constitutive response over the investigated states; we do not assume that microscopic polyconvexity guarantees a polyconvex homogenized energy.

2.4 Consistent constitutive tangent

The tangent must account for the change in microscopic equilibrium as the input strain changes. To make this dependence explicit, write the stress evaluator before substitution of the equilibrium state as $\widehat{\mathbf{s}}(\mathbf{a}, \mathbf{e})$, so that $\mathbf{s}(\mathbf{e}) = \widehat{\mathbf{s}}(\mathbf{a}^*(\mathbf{e}), \mathbf{e})$. If $\mathbf{r}_{,\mathbf{a}}$ is nonsingular on the selected branch, implicit differentiation of Eq. (8) gives

$$\frac{d\mathbf{a}^*}{d\mathbf{e}} = -\mathbf{r}_{,\mathbf{a}}^{-1}\mathbf{r}_{,\mathbf{e}}, \quad \mathbf{D} = \widehat{\mathbf{s}}_{,\mathbf{e}} - \widehat{\mathbf{s}}_{,\mathbf{a}}\mathbf{r}_{,\mathbf{a}}^{-1}\mathbf{r}_{,\mathbf{e}}. \quad (11)$$

All partial derivatives are evaluated at equilibrium, holding the other argument fixed. The first term differentiates the stress at fixed fluctuation coordinates; the second accounts for microscopic relaxation. Omitting that correction generally does not give the derivative of the deployed constitutive response. For the smooth hyperelastic branch, the result is also $\partial_{\mathbf{e}\mathbf{e}}^2 W$.

No explicit matrix inverse is needed in Eq. (11): the sensitivities follow from linear systems with the converged microscopic tangent. The reference implementation uses one nonlinear equilibrium solution followed by three sensitivity right-hand sides, one per strain component. Independent finite-difference checks perturb the input and re-solve equilibrium. Such checks assess the constitutive linearization, not the physical accuracy of the homogenization assumptions. The same consistency requirement applies to each surrogate: its tangent must differentiate its own stress evaluation.

2.5 Macroscopic equilibrium and the constitutive interface

The pair $(\mathbf{s}, \mathbf{D})$ is the common interface through which every constitutive evaluator enters the structural problem. Let $\mathbf{U}$ collect the free macroscopic nodal displacements, and let $\mathbf{e}_g(\mathbf{U})$ be the engineering Green strain at integration point g . With $\mathbf{B}_g = \partial \mathbf{e}_g / \partial \mathbf{U}$, reference-volume integration weights Ω_g , and dead external loads $\mathbf{f}_{\text{ext}}$, equilibrium reads

$$\mathbf{R}_M(\mathbf{U}) = \sum_g \Omega_g \mathbf{B}_g^T \mathbf{s}_g(\mathbf{e}_g(\mathbf{U})) - \mathbf{f}_{\text{ext}} = \mathbf{0}. \quad (12)$$

Its tangent contains material and geometric contributions,

$$\mathbf{K}_M = \sum_g \Omega_g \left[\mathbf{B}_g^T \mathbf{D}_g \mathbf{B}_g + \sum_{\alpha=1}^3 (s_g)_\alpha \frac{\partial^2 (e_g)_\alpha}{\partial \mathbf{U} \partial \mathbf{U}} \right]. \quad (13)$$

The first term uses the constitutive tangent; the second arises because the Green strain is nonlinear in the displacements. A structural Newton solver therefore needs both $\mathbf{s}_g$ and $\mathbf{D}_g$, irrespective of how the constitutive response is computed.

The nested FOM-FE² evaluation provides the reference instance of this interface: every constitutive query solves the full periodic RVE problem defined above. Section 3 next constructs effective energies that return $\mathbf{s}$ and $\mathbf{D}$ by differentiation and therefore remove the microscopic solve from the online stage. Section 4 then introduces the intrusive reduced references. Its iterative HPROM variants retain microscopic equilibrium in reduced coordinates and replace full assembly by hyperreduced integration, whereas the direct decoder bypasses that equilibrium solve. Despite these different routes, all models supply the same interface while the macroscopic mesh, loading, and equilibrium algorithm remain unchanged.

Holding that structural problem fixed makes the nested FOM-FE² solution the reference for measuring surrogate errors at fixed macro and micro discretizations. It is not a direct numerical simulation resolving every pore in the coupon. Homogenization error, boundary-layer effects, and mesh-discretization error are therefore distinct from the surrogate errors reported below. The cell itself is an ideal periodic material model, without a claim of statistical representativeness for a random microstructure. Accordingly, FE² denotes the nested calculation in which each constitutive query solves an RVE problem; a direct learned constitutive evaluation supplies the same interface but is not itself a microscopic finite element solve.

3 Learnable paired-feature polyconvex energies

The non-intrusive route developed here replaces the repeated online solution of the microscopic problem by a learned effective stored energy. Its central constitutive relation is

$$\mathbf{s}(\mathbf{e}) = \nabla_{\mathbf{e}} W(\mathbf{e}), \quad (14)$$

where $\mathbf{s}$ is the stress work-conjugate to the engineering-strain vector $\mathbf{e}$. Thus all stress components derive from a common potential, and the Hessian of W supplies a major-symmetric material tangent. Fitting stress

components independently does not, in general, enforce the cross-derivative identities required by such a potential [12]. This observation motivates the energy-based formulation adopted throughout this section.

One could stop at this point and construct a flexible hyperelastic network without restricting the curvature of its learned potential. We use precisely this option as the *Unconstrained energy* reference. It takes objective strain measures as inputs and obtains stress through Eq. (14). Beyond potential structure and objectivity, the only mechanical conditions imposed by construction are the reference conditions $W(\mathbf{I}) = 0$ and $\partial_e W(\mathbf{0}) = \mathbf{0}$. The first fixes the arbitrary additive energy constant; the second is enforced by an affine correction that cancels the stress at the undeformed state, following the consistency construction of As’ad et al. [12].¹

The Unconstrained energy model shows what can be represented before stronger finite-strain restrictions are introduced. Potential structure and reference normalization alone do not control deformation-gradient curvature or prevent finite energy as $J \rightarrow 0^+$. We therefore pursue polyconvexity by constructing the energy as a convex function of independent deformation minors, together with analytic terms that control the reference state and growth. The learned contribution combines a common family of paired directional features with one of two monotone convex cores: an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN). These cores provide alternative representations while preserving the same feature construction and mechanical guarantees.

The remainder of this section makes these distinctions precise. Figure 1 summarizes the construction, tracing the path from RVE data through the paired directional features and either convex core to the analytic completion of the stored energy, whose derivatives supply stress and tangent. We first state the mechanical requirements and separate potential structure from finite-strain stability. We then define the reference model, construct the paired features and convex cores, and establish the properties of the complete energy. Throughout, $\mathbf{F} \in \text{GL}^+(2)$, $J = \det \mathbf{F} > 0$, and derivatives with respect to deformation are taken with the learned parameters fixed.

3.1 Admissibility criteria and scope of the guarantees

A small prediction error is not sufficient to define an admissible hyperelastic surrogate. We therefore separate six requirements relevant to the present construction. The requirements themselves are standard in constitutive PANN formulations [13, 14]; their grouping and labels below are specific to this paper:

- (C1) *Potential structure*: stress is the derivative of a single-valued stored energy in work-conjugate variables.
- (C2) *Objectivity*: a superposed rigid rotation does not change that energy.
- (C3) *Stress symmetry and anisotropy*: the second Piola stress is symmetric, while directional dependence is allowed without imposing an unverified material symmetry group.
- (C4) *Reference normalization*: the undeformed state has zero energy and zero stress.
- (C5) *Polyconvexity*: the energy admits a convex representation in deformation minors, providing a sufficient rank-one convexity condition.
- (C6) *Growth*: energy diverges under volume collapse, unbounded volume expansion, and unbounded distortion at bounded positive volume.

This list is an organizing device, not a universal checklist or a claim that polyconvexity is necessary for every physical material. The Unconstrained energy reference satisfies (C1)–(C4) but carries no construction-level claim of (C5) or (C6). The ICNN and ICKAN models proposed here satisfy all six criteria by construction for every finite parameter set that obeys the stated architectural constraints.²

These statements rest on different levels of evidence. Constitutive identities follow from the representation itself; sufficient bounds hold under their stated assumptions; and sampled numerical checks verify an implementation but do not replace a proof. The discussion below maintains this distinction while examining each criterion.

Potential structure (C1). To ensure mutually compatible stresses, we derive all stress components from a single scalar potential. Equation (14) gives the resulting work-conjugate stress. It also implies that the work around every closed strain path Γ vanishes:

$$\oint_{\Gamma} \mathbf{s} \cdot d\mathbf{e} = 0. \quad (15)$$

¹No convexity or growth condition is imposed on this reference.

²Global nonnegative energy is not part of (C1)–(C6); the saved checkpoints are audited separately in Section 5.3.3 using the sufficient bound in Appendix C.

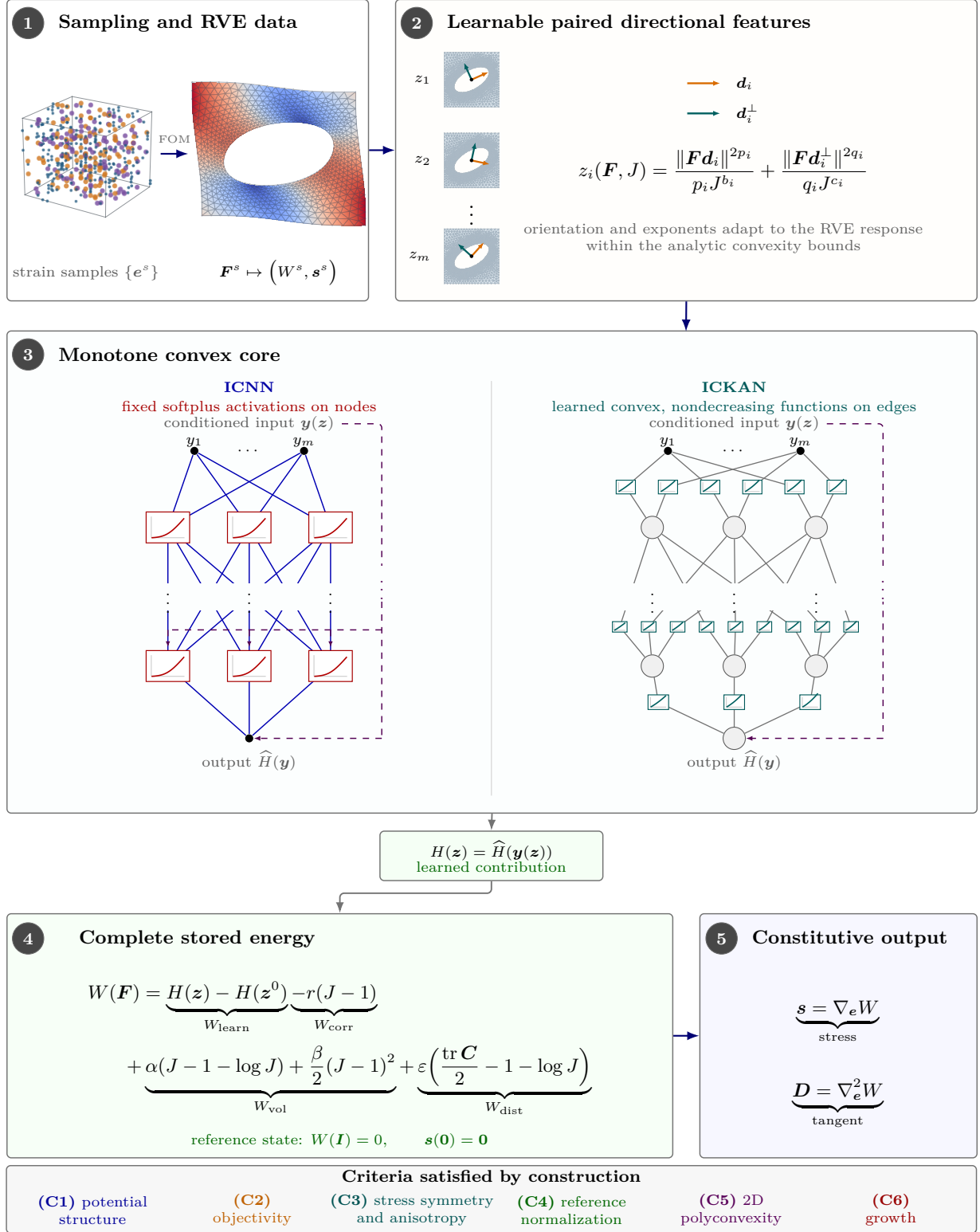


Figure 1: Overview of the proposed constitutive construction. Sampled strain states are evaluated with the periodic RVE model to supply energy and stress data. These responses train learnable paired directional features and a monotone convex ICNN or ICKAN core. Analytic reference and growth terms complete the stored energy. Differentiation then supplies the work-conjugate stress and material tangent. The stated properties follow from the complete construction rather than from sampled numerical checks.

For the isothermal purely elastic setting considered here, (C1) likewise gives $\mathbf{s} \cdot \dot{\mathbf{e}} - \dot{W} = 0$: the mechanical dissipation vanishes. Moreover, the stress Jacobian is the Hessian of W and therefore obeys the cross-derivative identities $\partial s_i / \partial e_j = \partial s_j / \partial e_i$. These statements follow from the scalar-potential representation itself, not from a small error at sampled states.

Objectivity and the three meanings of symmetry (C2)–(C3). Objectivity must remove dependence on the spatial observer without inadvertently imposing material isotropy. For a superposed proper rotation $\mathbf{Q}$, it requires

$$W(\mathbf{Q}\mathbf{F}) = W(\mathbf{F}), \quad \mathbf{P}(\mathbf{Q}\mathbf{F}) = \mathbf{Q}\mathbf{P}(\mathbf{F}). \quad (16)$$

Using $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ as an energy input enforces the first identity, because a superposed spatial rotation leaves $\mathbf{C}$ unchanged; differentiating the objective energy then gives the stress transformation. Objectivity does not itself imply either potential structure or material isotropy. For the energy models considered here, the symmetry $\mathbf{S} = \mathbf{S}^T$ follows from differentiation with respect to symmetric strain. It implies a symmetric Cauchy stress $\boldsymbol{\sigma} = J^{-1} \mathbf{F} \mathbf{S} \mathbf{F}^T$, as required by angular-momentum balance in the present continuum.

Objectivity still permits anisotropy. Material symmetry is a different invariance: a right action $\mathbf{F} \mapsto \mathbf{F} \mathbf{Q}_m$ leaves the energy unchanged only when $\mathbf{Q}_m$ belongs to the material symmetry group. Section 3.3 will implement this distinction using directions attached to the reference material; a superposed spatial rotation acts on $\mathbf{F}$ from the left and does not alter the directional stretch measured in that reference. Finally, the constitutive tangent $\mathbf{D}$ of Eq. (3) measures how the three stress components change with engineering strain. For an energy-based law it is the Hessian $\partial_{\mathbf{e}\mathbf{e}}^2 W$. Its *major symmetry*, $\mathbf{D} = \mathbf{D}^T$, is a third property: it follows from a twice differentiable scalar energy, not merely from a symmetric stress tensor.

Reference normalization (C4). To make the learned energy consistent with the undeformed reference state, we require both zero energy and zero stress:

$$W(\mathbf{I}) = 0, \quad \mathbf{s}(\mathbf{0}) = \mathbf{0}, \quad (17)$$

where $\mathbf{s}(\mathbf{0}) = \mathbf{0}$ is equivalent to $\mathbf{P}(\mathbf{I}) = \mathbf{0}$ in the undeformed state. The first condition fixes an arbitrary additive energy constant. The second is a substantive stress condition, and the admissible correction depends on the model class. Section 3.2 subtracts the raw reference value and a term linear in $\mathbf{E}$ for the Unconstrained energy reference. That correction preserves its scalar potential, but an arbitrary linear function of $\mathbf{E}$ need not preserve polyconvexity in $\mathbf{F}$. Section 3.5 therefore uses a different correction for the proposed models: it is affine in independent deformation minors and thus preserves convexity in those variables.

Polyconvexity and finite-strain stability (C5). Potential structure alone does not control the energy curvature under changes in deformation. Direct convexity in $\mathbf{F}$ is generally too restrictive for objective finite-strain energies with singular-volume growth. Polyconvexity instead imposes convexity in independent deformation minors [17, 18]. In two dimensions,³ this means that the energy admits the representation

$$W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F}), \quad \mathcal{P} \text{ convex in its independent arguments.} \quad (18)$$

Here $\mathcal{P}$ is defined on the convex set $\mathbb{R}^{2 \times 2} \times (0, \infty)$, treating its matrix and scalar arguments as independent; only its evaluation in W sets the second argument equal to $\det \mathbf{F}$. The cofactor is linear in the entries of a 2×2 matrix, so it need not be introduced as an additional independent argument. Convexity of $\mathcal{P}$ implies that W cannot curve downward along an admissible rank-one perturbation of $\mathbf{F}$. Appendix A derives this consequence and states its differential form.

To distinguish the present construction from alternatives that impose convexity in strain space [12, 15], we emphasize that convexity in $\mathbf{C}$ or $\mathbf{E}$ does not imply polyconvexity. The reason is that $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$ depends quadratically on $\mathbf{F}$; Appendix A provides the corresponding derivation and an analytical counterexample.

Growth and the limits of the guarantees (C6). Polyconvexity alone does not prevent volume collapse or unbounded deformation. We therefore require $W \rightarrow \infty$ as $J \rightarrow 0^+$, as $J \rightarrow \infty$, and as $\|\mathbf{F}\| \rightarrow \infty$ while J remains bounded away from zero and infinity. These growth conditions contribute to coercivity but do not alone prove existence of a minimizer: the standard existence argument also requires admissible boundary data and finite-energy competitors [18]. Nor does a constitutive barrier guarantee convergence of an unconstrained Newton iteration. Architectural guarantees and numerical solver behavior are therefore reported separately.

³The standard three-dimensional definition uses $(\mathbf{F}, \text{cof } \mathbf{F}, \det \mathbf{F})$. The two-dimensional guarantee established here does not imply an admissible extension to arbitrary out-of-plane deformation.

3.2 Unconstrained energy reference

The most flexible energy model used in the comparison is a smooth neural network $\widehat{W}_\theta$ acting on

$$\mathbf{x}(\mathbf{e}) = (C_{11} - 1, C_{22} - 1, C_{12}, J - 1)^T. \quad (19)$$

Here $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ and $J = \sqrt{\det \mathbf{C}}$, so the four objective inputs are functions of the same three engineering-strain components; they do not introduce an additional state variable. Denote the resulting raw energy by $\widetilde{W}(\mathbf{e}) = \widehat{W}_\theta(\mathbf{x}(\mathbf{e}))$. A generic network does not satisfy the reference conditions automatically, so we define the normalized energy as

$$W_{\text{unc}}(\mathbf{e}) = \widetilde{W}(\mathbf{e}) - \widetilde{W}(\mathbf{0}) - \partial_{\mathbf{e}} \widetilde{W}(\mathbf{0}) \cdot \mathbf{e}. \quad (20)$$

At $\mathbf{e} = \mathbf{0}$, the first two terms cancel and the linear term vanishes, giving $W_{\text{unc}}(\mathbf{0}) = 0$. Differentiation gives $\partial_{\mathbf{e}} W_{\text{unc}}(\mathbf{e}) = \partial_{\mathbf{e}} \widetilde{W}(\mathbf{e}) - \partial_{\mathbf{e}} \widetilde{W}(\mathbf{0})$, and therefore zero reference stress. Because the correction is affine in $\mathbf{e}$, it does not change the Hessian of the raw network. Equation (20) thus fixes the reference energy and stress while leaving the learned curvature unrestricted.

Using components of $\mathbf{C}$ in a material reference basis ensures objectivity under superposed spatial rotations but does not impose isotropy: the network may depend differently on C_{11} , C_{22} , and C_{12} . A change of material reference axes must transform these tensor components and the represented law consistently. This model is used only as the flexible energy-based reference for interpreting the constrained ICNN and ICKAN results.

3.3 Feature design: expressiveness and admissibility

The first design question is how to describe deformation without erasing the effective anisotropy of the RVE. Classical invariants provide the natural starting point. For a three-dimensional right Cauchy–Green tensor $\mathbf{C}_3$, the familiar first, second, and third principal invariants are

$$I_1^{(3)} = \text{tr } \mathbf{C}_3, \quad I_2^{(3)} = \frac{1}{2}[(\text{tr } \mathbf{C}_3)^2 - \text{tr}(\mathbf{C}_3^2)], \quad I_3^{(3)} = \det \mathbf{C}_3. \quad (21)$$

They describe principal stretches without recording their orientation relative to the material. An energy depending only on these scalars is therefore isotropic. In the present two-dimensional setting, the corresponding two independent principal invariants are $I_1 = \text{tr } \mathbf{C}$ and $\det \mathbf{C} = J^2$; a third independent principal invariant is not available.

To retain directional information, one can supplement the isotropic measures with a structural direction $\mathbf{d}$. The familiar anisotropic stretch invariant is

$$I_4(\mathbf{d}) = \mathbf{d}^T \mathbf{C} \mathbf{d} = \mathbf{C} : (\mathbf{d} \otimes \mathbf{d}) = \|\mathbf{F} \mathbf{d}\|^2, \quad \|\mathbf{d}\| = 1. \quad (22)$$

It measures squared stretch along a material direction and remains unchanged by a superposed spatial rotation. Such directional invariants are standard ingredients of anisotropic hyperelastic energies [40, 41]. Here $\mathbf{d}$ need not represent a physical fiber: learned directions provide objective scalar descriptors of the RVE’s effective anisotropy. In two dimensions, $\mathbf{d}$ is naturally accompanied by its perpendicular $\mathbf{d}^\perp$. For this orthonormal pair,

$$I_4(\mathbf{d}) + I_4(\mathbf{d}^\perp) = I_1, \quad \|\text{cof } \mathbf{F} \mathbf{d}\|^2 = I_4(\mathbf{d}^\perp). \quad (23)$$

The first identity shows that the pair resolves how I_1 is distributed between two perpendicular directions: their sum recovers I_1 , while their individual values retain directional information. The second shows that a cofactor-based directional measure reads the stretch in the perpendicular direction rather than introducing an independent third in-plane descriptor. These observations motivate the paired directional building block used below.

Objective directional descriptors solve only the first part of the design problem. Because the downstream core is constrained to be nondecreasing in every input, the feature coordinates also determine which deformation states the core must order. The feature map must therefore be designed together with that monotonicity constraint. Consider a feature vector $\mathbf{z}(\mathbf{F})$ and a scalar core H nondecreasing in each input. For any two deformation states,

$$\mathbf{z}(\mathbf{F}_B) \geq \mathbf{z}(\mathbf{F}_A) \text{ componentwise} \implies H(\mathbf{z}(\mathbf{F}_B)) \geq H(\mathbf{z}(\mathbf{F}_A)). \quad (24)$$

This implication holds regardless of the network’s width. If the complete constitutive energy is $W = H(\mathbf{z}) + W_{\text{corr}}$, where W_{corr} collects the analytic normalization and growth terms, the ordering constrains the neural contribution $W - W_{\text{corr}}$ rather than the complete energy. The feature map and analytic terms must therefore be considered together.

This observation explains why the choice of an admissible feature construction can matter more than the capacity of its convex core. Changing the number of features does not change the number of independent strains: the features remain functions of the same three engineering-strain components. They embed that state into variables on which the network is constrained to be monotone and convex. Additional features can enlarge the admissible response family, but their number alone does not prove completeness.

We therefore seek features that retain directional stretch, reshape the componentwise ordering seen by the monotone core, and remain convex in the independent variables $(\mathbf{F}, J)$. The paired construction below achieves these three objectives by coupling directional stretch to volume through admissible determinant exponents. A feature can consequently decrease along a path when its denominator grows sufficiently, even if a numerator increases. Its reference derivative is also controlled analytically, which will allow the reference stress to be removed without sacrificing convexity.

3.3.1 Paired directional features

Let m denote the number of scalar features. For each $i = 1, \dots, m$, let $\mathbf{d}_i = (\cos \theta_i, \sin \theta_i)^T$ and $\mathbf{d}_i^\perp = (-\sin \theta_i, \cos \theta_i)^T$. Define

$$\begin{aligned} z_i(\mathbf{F}, J) &= \frac{I_4(\mathbf{d}_i)^{p_i}}{p_i J^{b_i}} + \frac{I_4(\mathbf{d}_i^\perp)^{q_i}}{q_i J^{c_i}} \\ &= \frac{\|\mathbf{F}\mathbf{d}_i\|^{2p_i}}{p_i J^{b_i}} + \frac{\|\mathbf{F}\mathbf{d}_i^\perp\|^{2q_i}}{q_i J^{c_i}}, \end{aligned} \quad (25)$$

where θ_i sets the orientation of the reference directions, p_i, q_i are the powers of the squared-stretch invariants, and b_i, c_i control each contribution's coupling to the area ratio J . These roles provide expressiveness, but the exponents cannot be chosen arbitrarily if the feature is to remain convex. They satisfy

$$p_i, q_i \geq \frac{1}{2}, \quad 0 \leq b_i \leq 2p_i - 1, \quad 0 \leq c_i \leq 2q_i - 1. \quad (26)$$

The lower bounds on p_i and q_i make the corresponding powers of the directional norms convex, while the upper bounds on b_i and c_i prevent their coupling with J from introducing a negative-curvature direction. Appendix B derives these conditions from the Hessian of a single stretch-determinant contribution. In the learned-feature variants, θ_i and the four exponent parameters are optimized, with p_i, q_i, b_i, c_i subject to Eq. (26); the fixed-feature controls of Section 5.2 hold the same initialized feature parameters constant while training the core. We use $\text{softplus}(x) = \log(1 + e^x)$ and $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$, whose ranges for $x \in \mathbb{R}$ are $(0, \infty)$ and $(0, 1)$, respectively. With unconstrained parameters $\hat{p}_i, \hat{q}_i, \hat{b}_i, \hat{c}_i \in \mathbb{R}$, set

$$\begin{aligned} p_i &= \frac{1}{2} + \text{softplus}(\hat{p}_i), & b_i &= (2p_i - 1) \text{sigmoid}(\hat{b}_i), \\ q_i &= \frac{1}{2} + \text{softplus}(\hat{q}_i), & c_i &= (2q_i - 1) \text{sigmoid}(\hat{c}_i). \end{aligned} \quad (27)$$

The angles require no admissibility bound: for any fixed θ_i , the maps $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i$ and $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i^\perp$ remain linear in $\mathbf{F}$. The transformations in Eq. (27) preserve the exponent inequalities during training, so a penalty loss is not needed to enforce them. For finite real values of the unconstrained parameters, the bounds in Eq. (26) hold strictly; boundary values are approached as limits. Each z_i is one scalar feature containing two perpendicular contributions. The two contributions need not have equal exponents. Consequently, pairing perpendicular directions does not impose invariance under their interchange.

The construction was also designed to simplify reference normalization. Its paired and scaled contributions make the reference strain gradient of every feature isotropic. Consequently, the neural contribution has an isotropic reference stress, which a single scalar determinant correction can cancel in Section 3.5. The following result gathers the objectivity, convexity, and reference-state properties needed there.

Proposition 1. *Under the constraints in Eq. (26), each feature in Eq. (25) is objective and convex in the independent variables $(\mathbf{F}, J)$ on $J > 0$. Its value and engineering-strain gradient at the identity, evaluated with $J = \det \mathbf{F}$, are*

$$z_i^0 = \frac{1}{p_i} + \frac{1}{q_i}, \quad \partial_{\mathbf{e}} z_i|_{\mathbf{0}} = \rho_i(1, 1, 0)^T, \quad \rho_i = 2 - \frac{b_i}{p_i} - \frac{c_i}{q_i}. \quad (28)$$

The proof is given in Appendix B. The mechanism behind the reference gradient is that $1/p_i$ and $1/q_i$ cancel the stretch-power multipliers upon differentiation. The two perpendicular directional contributions then sum to $2\mathbf{I}$, while the determinant terms contribute $-(b_i/p_i + c_i/q_i)\mathbf{I}$. Thus ρ_i is the scalar amplitude of an isotropic reference gradient: its two normal components are equal and its shear component is zero. This reference property does not imply isotropy of the response away from the reference state.

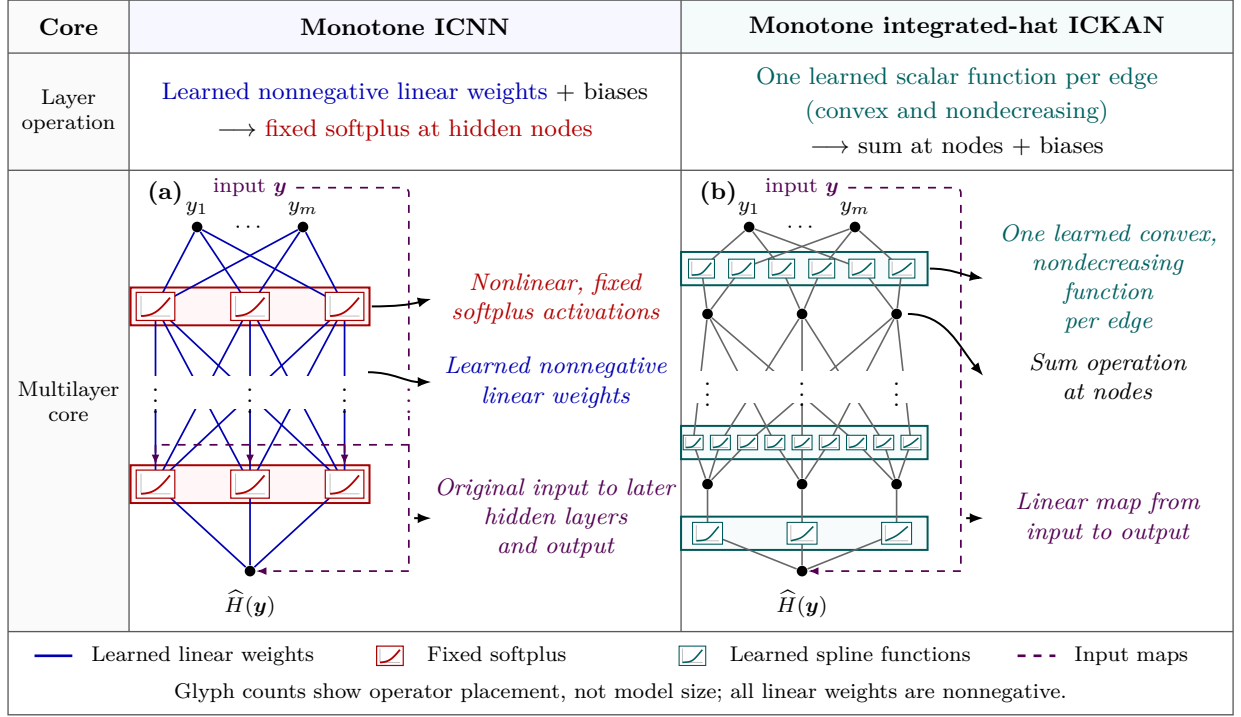


Figure 2: Multilayer monotone cores mapping the conditioned paired features $\mathbf{y}$ to the scalar contribution $\hat{H}(\mathbf{y})$. (a) The ICNN learns nonnegative linear maps and applies fixed softplus activations at hidden nodes. (b) The ICKAN learns one convex, nondecreasing scalar function per edge and sums them at nodes. Purple dashed paths denote nonnegative maps of the complete input; ellipses denote intervening layers. Glyph counts indicate where each operation acts, not deployed widths or parameter counts. Unrestricted biases are omitted, and the diagram represents the neural cores rather than the complete energy.

The raw features are now mechanically admissible, but their numerical magnitudes may differ. Before passing them to the core, we therefore use $y_i = (z_i - z_i^{\text{cen}})/s_i$, where z_i^{cen} is a centering value and $s_i > 0$ is a scaling factor. Both are independent of deformation for any fixed parameter set. This positive affine conditioning preserves both convexity in independent $(\mathbf{F}, \mathbf{J})$ and the componentwise ordering required by the monotone core. The choices used in the constitutive assessments are specified in Section 5.1.1.

This feature design has a precise but limited purpose: it supplies objective, expressive, and admissible inputs to the constrained core. Learning the directions does not amount to identifying or enforcing a complete material symmetry group, and no completeness or universal-approximation theorem is asserted for a finite collection of these features followed by a monotone convex core.

3.4 Monotone ICNN and integrated-hat ICKAN cores

The feature construction of Section 3.3 resolves the input side of the constitutive design: each z_i is objective and convex in the independent variables $(\mathbf{F}, \mathbf{J})$. The remaining question is how a learned core may combine these inputs without undoing that convexity. The core receives the conditioned vector $\mathbf{y} = (y_1, \dots, y_m)^T$, with $y_i = (z_i - z_i^{\text{cen}})/s_i$. Because $s_i > 0$, this affine conditioning preserves the convexity and componentwise ordering of the physical features.

A convex core alone is not sufficient. Composition with the convex feature map preserves convexity only when the core is also componentwise nondecreasing [13, 14]. We enforce these same two properties in two different ways. The ICNN learns nonnegative linear maps and applies fixed softplus activations at its hidden nodes. The ICKAN instead learns a convex, nondecreasing scalar function on every edge and sums the incoming contributions at each node. The following definitions make these two mechanisms precise; Figure 2 then compares their multilayer architectures.

3.4.1 Monotone ICNN

Let L denote the number of hidden layers and $\mathbf{a}^{(\ell)}$ the vector of activations in layer ℓ , for $\ell = 1, \dots, L$. We write $\hat{H}(\mathbf{y})$ for the scalar neural core acting on the conditioned features $\mathbf{y}$; the complete energy additionally

includes the reference-normalization and growth terms of Section 3.5. The deployed ICNN has the form

$$\begin{aligned} \mathbf{a}^{(1)} &= \phi(\mathbf{U}_1 \mathbf{y} + \mathbf{b}_1), \\ \mathbf{a}^{(\ell)} &= \phi(\mathbf{W}_\ell \mathbf{a}^{(\ell-1)} + \mathbf{U}_\ell \mathbf{y} + \mathbf{b}_\ell), \quad \ell = 2, \dots, L, \\ \hat{H}(\mathbf{y}) &= \mathbf{v}^T \mathbf{a}^{(L)} + \mathbf{w}^T \mathbf{y} + b_{\text{out}}, \end{aligned} \quad (29)$$

where $\phi = \text{softplus}$ is applied componentwise. Its derivatives satisfy $\phi'(x) = \text{sigmoid}(x) > 0$ and $\phi''(x) = \text{sigmoid}(x)[1 - \text{sigmoid}(x)] > 0$. Thus the activation is smooth, increasing, and convex; its smoothness also makes the core C^2 , as required for a continuous constitutive tangent. The matrices $\mathbf{U}_\ell$ and $\mathbf{W}_\ell$ contain the weights from $\mathbf{y}$ and the preceding hidden layer, respectively, and $\mathbf{b}_\ell$ is the layer bias vector. The output weight vectors $\mathbf{v}$ and $\mathbf{w}$ act on $\mathbf{a}^{(L)}$ and $\mathbf{y}$, respectively; b_{out} is an additive scalar output bias.

Every entry of $\mathbf{W}_\ell, \mathbf{U}_\ell, \mathbf{v}, \mathbf{w}$ is nonnegative, enforced by softplus transformations of unconstrained weight parameters. These signs play two related but distinct roles. The matrices $\mathbf{W}_\ell$ and the output vector $\mathbf{v}$ multiply convex hidden activations, so their nonnegativity preserves convexity as well as monotonicity. The maps $\mathbf{U}_\ell \mathbf{y}$ and $\mathbf{w}^T \mathbf{y}$ are affine and therefore do not affect convexity for either sign, but nonnegative $\mathbf{U}_\ell$ and $\mathbf{w}$ are required for componentwise monotonicity. Biases are unrestricted because adding constants to a preactivation preserves both properties. Hidden biases can change the core's derivatives with respect to $\mathbf{y}$, whereas b_{out} only shifts $\hat{H}$ and leaves its derivatives unchanged.

The preservation argument proceeds layer by layer. The first hidden layer is convex because softplus is convex and its argument is affine, and it is nondecreasing because $\mathbf{U}_1$ is nonnegative. Assuming that $\mathbf{a}^{(\ell-1)}$ is convex and nondecreasing, the nonnegative map $\mathbf{W}_\ell \mathbf{a}^{(\ell-1)}$, the nonnegative affine input map, and composition with ϕ preserve both properties in layer ℓ . The same argument applies to the output, proving by induction that $\hat{H}$ is convex and componentwise nondecreasing. This is the ICNN composition argument of Amos et al. [42], supplemented here by nonnegative direct-input maps to guarantee monotonicity.

3.4.2 Monotone integrated-hat ICKAN

Unlike the ICNN, which applies a fixed activation after an affine weighted sum at each hidden node, the ICKAN assigns a learned scalar function to each connection (edge) between layers. Each node sums the transformed contributions from its incoming connections. To make the relation between these edge functions and the scalar core explicit, set $\mathbf{h}^{(0)} = \mathbf{y}$ and let n_ℓ be the width of layer ℓ , with $n_0 = m$. The hidden layers and scalar output are

$$\begin{aligned} h_j^{(\ell)} &= \sum_{i=1}^{n_{\ell-1}} f_{ji}^{(\ell)} \left(h_i^{(\ell-1)} \right) + b_j^{(\ell)}, \quad j = 1, \dots, n_\ell, \quad \ell = 1, \dots, L, \\ \hat{H}(\mathbf{y}) &= \sum_{i=1}^{n_L} f_i^{\text{out}} \left(h_i^{(L)} \right) + \mathbf{w}^T \mathbf{y} + b_{\text{out}}, \quad \mathbf{w} \geq \mathbf{0}. \end{aligned} \quad (30)$$

Thus $f_{ji}^{(\ell)}$ transforms one scalar carried by an edge, $h_j^{(\ell)}$ is the sum formed at a node, and $\hat{H}(\mathbf{y})$ is the complete ICKAN core. The nonnegative linear skip $\mathbf{w}^T \mathbf{y}$ connects the conditioned input directly to the scalar output. It remains to specify the functions appearing on the edges. For any $f_{ji}^{(\ell)}$ or f_i^{out} in Eq. (30), let x denote its scalar input. Each edge uses the common form

$$f(x) = ax + \sum_k c_k B_k(x), \quad a, c_k \geq 0, \quad (31)$$

where a is the learned linear slope, c_k are learned weights of the spline bases B_k , and k indexes these bases. Softplus transformations of unconstrained parameters impose the signs of a and c_k . Additive constants are represented once, through the unrestricted node biases in Eq. (30), rather than redundantly on every incoming edge.

The remaining design task is to learn the shape of each edge without allowing negative slope or curvature. Rather than check these inequalities at sampled inputs after training, we parameterize the edge curvature directly. Each B_k'' is a nonnegative triangular hat centered at a knot κ_k . The knots form a fixed uniform grid with spacing $h > 0$; neither their locations nor the spacing are optimized. Writing $s = (x - \kappa_k)/h$, one curvature basis is

$$B_k''(x) = \max(1 - |s|, 0). \quad (32)$$

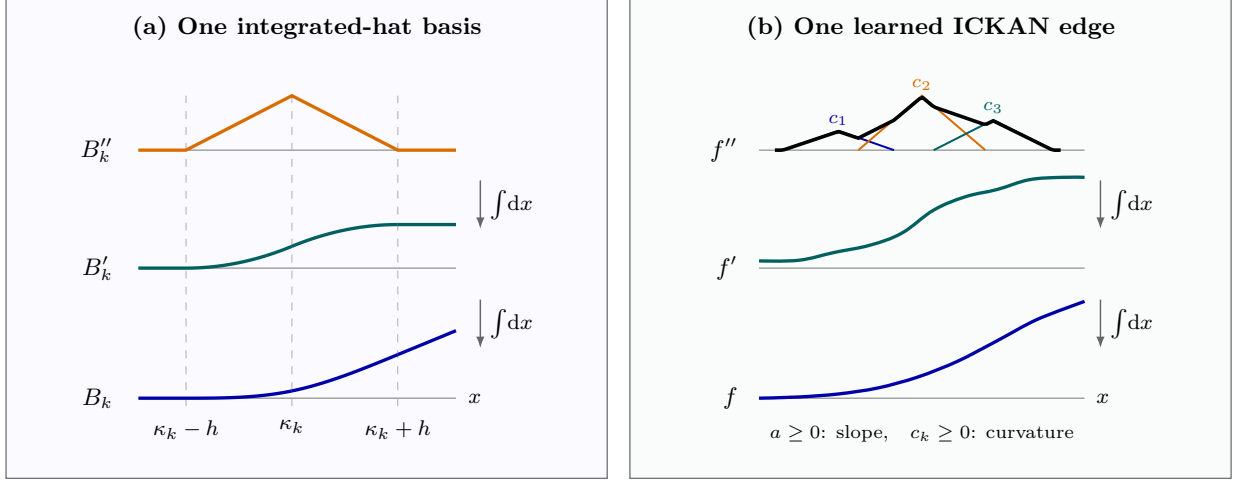


Figure 3: Integrated-hat parameterization of an ICKAN edge. (a) A localized nonnegative curvature hat integrates to a nonnegative slope and a convex C^2 basis with a zero left tail and a linear right tail. (b) Nonnegative learned coefficients combine several fixed bases into a nonnegative edge curvature; integration gives a nondecreasing slope and a convex, nondecreasing connection. The curves illustrate the analytic construction; they do not represent the deployed grid or parameter count.

Integrating twice from the left conditions $B_k(x) = B'_k(x) = 0$ for $x \leq \kappa_k - h$ gives

$$B_k(x) = h^2 \begin{cases} 0, & s \leq -1, \\ (s+1)^3/6, & -1 < s \leq 0, \\ (-s^3 + 3s^2 + 3s + 1)/6, & 0 < s < 1, \\ s, & s \geq 1. \end{cases} \quad (33)$$

Figure 3 shows both levels of the construction: integration of one curvature basis and assembly of a learned edge from several bases.

The value and first two derivatives in Eq. (33) match at every join, and B_k'' vanishes at the outer joins. Thus B_k is globally C^2 . Its curvature is localized to $[\kappa_k - h, \kappa_k + h]$, but the integrated basis is not: it has a zero left tail and a linear right tail. Moreover, $B'_k(-\infty) = 0$ and $B_k'' \geq 0$ imply $B'_k \geq 0$. The complete edge therefore satisfies

$$f''(x) = \sum_k c_k B_k''(x) \geq 0, \quad f'(x) = a + \sum_k c_k B'_k(x) \geq 0, \quad (34)$$

for every learned parameter set with $a, c_k \geq 0$. Hence each edge is convex and nondecreasing by construction, rather than by a sampled numerical test.

Equation (30) sums these scalar edge functions at each node. Because every edge is convex and nondecreasing, induction across the layers shows that every $h_j^{(\ell)}$, and finally $\hat{H}$, is convex and componentwise nondecreasing in $\mathbf{y}$. The nonnegative input-to-output skip preserves both properties. The fixed knot grid and finite number of bases limit where an individual edge can learn curvature: outside their combined support, the edge has linear tails. These tails are linear in the scalar input x , not in $\mathbf{F}$; the complete energy still contains nonlinear features and analytic growth terms. The grid used in the constitutive assessment is reported in Section 5.1.1. This core is inspired by the monotonic input-convex KAN framework [48], with the integrated-hat bases and analytic tails specified here.

3.5 The complete energy and its admissibility guarantees

Sections 3.3 and 3.4 constructed the physical feature vector $\mathbf{z}$, its conditioned representation $\mathbf{y}(\mathbf{z})$, and the scalar neural core $\hat{H}(\mathbf{y})$. We write

$$H(\mathbf{z}) = \hat{H}(\mathbf{y}(\mathbf{z})) \quad (35)$$

for the resulting learned contribution. This term represents the anisotropic response, but it does not by itself set the energy and stress references or prescribe growth at the boundary of the admissible deformation domain.

To complete the construction, we first quantify the residual stress produced by H at the undeformed reference state. Let $\mathbf{z}^0 = \mathbf{z}(\mathbf{I}, 1)$. With the feature-gradient factors ρ_i from Eq. (28), define the core

sensitivities and the scalar amplitude of this residual stress as

$$h_i = \partial_{z_i} H(\mathbf{z}^0) \geq 0, \quad r = \sum_i h_i \rho_i. \quad (36)$$

For a fixed parameter set, $H(\mathbf{z}^0)$ and r are constants with respect to deformation. The complete stored energy then combines the learned contribution with analytic reference-normalization and growth terms:

$$\begin{aligned} W(\mathbf{F}) = & \underbrace{H(\mathbf{z}) - H(\mathbf{z}^0)}_{(i) \, W_{\text{learn}}} + \underbrace{[-r(J-1)]}_{(ii) \, W_{\text{corr}}} \\ & + \underbrace{\alpha(J-1 - \log J) + \frac{\beta}{2}(J-1)^2}_{(iii) \, W_{\text{vol}}} \\ & + \underbrace{\varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right)}_{(iv) \, W_{\text{dist}}}, \quad \alpha, \beta, \varepsilon > 0. \end{aligned} \quad (37)$$

Here $J = \det \mathbf{F}$, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and $\mathbf{z} = \mathbf{z}(\mathbf{F}, J)$. The positive energy-density coefficients α , β , and ε weight the analytic growth terms. For either core, the composition in Eq. (35) is twice continuously differentiable, convex, and componentwise nondecreasing in $\mathbf{z}$. The underbraces name the four contributions used below: W_{learn} carries the learned anisotropic response, while W_{corr} , W_{vol} , and W_{dist} impose the reference and growth behavior.

Each contribution solves a distinct constitutive problem:

- (i) **Learned response and energy reference (W_{learn}).** At $\mathbf{F} = \mathbf{I}$, the features equal $\mathbf{z}^0$, and hence

$$W(\mathbf{I}) = H(\mathbf{z}^0) - H(\mathbf{z}^0) = 0. \quad (38)$$

The remaining terms also vanish because $J = 1$ and $\text{tr } \mathbf{C} = 2$. Since $H(\mathbf{z}^0)$ is constant with respect to deformation, its subtraction fixes the energy reference without changing the stress or tangent.

- (ii) **Reference-stress correction (W_{corr}).** A constant subtraction cannot change stress. By the chain rule and Eq. (28), the learned contribution instead has the reference value

$$\nabla_{\mathbf{e}} H(\mathbf{z})|_0 = \sum_i h_i \nabla_{\mathbf{e}} z_i|_0 = \sum_i h_i \rho_i (1, 1, 0)^T = r(1, 1, 0)^T. \quad (39)$$

The identity $J = \sqrt{(1 + 2E_{11})(1 + 2E_{22}) - (2E_{12})^2}$ gives $\nabla_{\mathbf{e}} J|_0 = (1, 1, 0)^T$. Holding r fixed when differentiating therefore yields

$$\nabla_{\mathbf{e}} [H(\mathbf{z}) - r(J-1)]|_0 = r(1, 1, 0)^T - r(1, 1, 0)^T = \mathbf{0}. \quad (40)$$

A single scalar r suffices because the paired features make only the *reference* stress isotropic; the learned response away from $\mathbf{F} = \mathbf{I}$ remains anisotropic. The correction is zero in value at $J = 1$ and, being affine in the independent determinant variable, preserves convexity for either sign of r . This affine determinant normalization follows the strategy of Linden et al. [14]; Eq. (28) makes it applicable to the present directional feature family.

- (iii) **Volumetric growth (W_{vol}).** Its two parts control opposite limits: $\alpha(J-1 - \log J)$ diverges as $J \rightarrow 0^+$, whereas $\beta(J-1)^2/2$ grows quadratically as $J \rightarrow \infty$. At the reference,

$$W_{\text{vol}}(1) = 0, \quad W'_{\text{vol}}(J) = \alpha(1 - J^{-1}) + \beta(J-1), \quad W'_{\text{vol}}(1) = 0. \quad (41)$$

Thus this term changes neither the reference energy nor the reference stress. Both parts are needed: the quadratic term alone would remain finite at collapse, with limit $\beta/2$.

- (iv) **Distortional growth at bounded area (W_{dist}).** A function of J alone cannot detect a change of shape that preserves area. For example, with $\lambda > 0$,

$$\mathbf{F} = \text{diag}(\lambda, \lambda^{-1}), \quad J = 1, \quad \varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right) = \frac{\varepsilon}{2} (\lambda^2 + \lambda^{-2} - 2). \quad (42)$$

This contribution diverges as $\lambda \rightarrow \infty$, although the area remains constant. More generally, $\text{tr } \mathbf{C} = \|\mathbf{F}\|_F^2$ supplies growth under unbounded deformation at bounded positive J . At the reference, the constant -1 sets its value to zero. Using $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ and $\partial_{\mathbf{E}} J|_0 = \mathbf{I}$, its stress contribution is

$$\partial_{\mathbf{E}} \left[\varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right) \right] \Big|_0 = \varepsilon [\mathbf{I} - J^{-1} \partial_{\mathbf{E}} J|_0] = \mathbf{0}. \quad (43)$$

Together, W_{learn} and W_{corr} set the learned response and its reference state, while W_{vol} and W_{dist} prescribe growth outside the sampled regime. Zero value and zero first derivative at the reference do not imply zero second derivative: the analytic terms still contribute to the reference stiffness because J is nonlinear in $\mathbf{e}$. The following proposition gathers the guarantees of their complete sum.

Proposition 2. *Let the features satisfy Eq. (26), let the core be either of the C^2 convex, componentwise nondecreasing constructions in Section 3.4, and let $\alpha, \beta, \varepsilon > 0$. Then Eq. (37) defines a C^2 , objective, two-dimensional polyconvex energy on $\text{GL}^+(2)$. Its associated second Piola stress $\mathbf{S}$ is symmetric, its work-conjugate representation is $\mathbf{s} = \nabla_{\mathbf{e}} W$, $W(\mathbf{I}) = 0$, $\mathbf{s}(\mathbf{0}) = \mathbf{0}$, and W diverges under collapse, unbounded expansion, and unbounded distortion at bounded positive area. Thus the construction satisfies (C1)–(C6).*

The proof is given in Appendix B.3. In brief, the convex, componentwise nondecreasing core preserves feature convexity; the remaining terms are constant, affine, or convex in independent $(\mathbf{F}, J)$. The reference identities and growth limits then complete the argument. These guarantees hold for every admissible parameter set and therefore do not depend on the training samples or fitting error. Predictive accuracy and structural deployment are assessed separately in Sections 5.3.2 and 5.3.5. Global nonnegative energy is not an architectural guarantee; it is audited for the saved checkpoints in Section 5.3.3 using the sufficient bound derived in Appendix C.

The three constitutive constructions carried into the numerical assessment are summarized in Table 1. The ICNN and ICKAN columns include both fixed- and learned-feature variants; learning the feature parameters changes the representation, not the guarantees established above.

Table 1: Constitutive constructions carried into the numerical assessment. Criteria (C1)–(C6) are defined in Section 3.1; guarantees refer to the specific constructions in this section.

	Unconstrained energy	ICNN	ICKAN
Role	Flexible hyperelastic reference	Guaranteed paired-feature energy	Guaranteed paired-feature energy
Inputs	Objective measures from $(\mathbf{C}, J)$	Paired features (fixed or learned)	Paired features (fixed or learned)
Learned scalar map	Smooth energy network	Nonnegative weights; fixed activations	Convex, nondecreasing edge functions
(C1)–(C4)	By construction	By construction	By construction
(C5)–(C6)	Not imposed	By construction	By construction

Section 4 next defines the intrusive microscopic references used in the structural accuracy–cost comparison. Section 5 then evaluates whether the constitutive implementations reproduce the constructions established above and how accurately the resulting energies approximate the RVE response.

4 Projection-based reduced-order models

The learned energies of Section 3 remove the microscopic solve from online constitutive evaluation. To place their structural accuracy and cost against physics-based alternatives, Section 5.3.5 also uses three projection-based microscopic surrogates as intrusive references. These models instead approximate the microscopic displacement field while retaining the underlying constituent law. Reduced coordinates decrease the number of equilibrium unknowns; hyperreduction decreases the required element evaluations. We consider an affine HPROM, an iterative HPROM–ANN with adaptive integration, and its direct counterpart D-HPROM–ANN.

Figure 4 provides a visual roadmap. Its first three stages follow the offline construction from RVE snapshots to the affine and nonlinear state representations; the final two compare the three online routes and their common stress–tangent output. The details are introduced in the remainder of this section.

4.1 Constrained microscopic reference

All three reduced evaluators derive from the periodic RVE problem of Section 2. Following the nomenclature of Ref. [24], we refer to its N -dimensional constrained discretization as the high-dimensional model (HDM), or full-order model (FOM). A PROM solves the same equilibrium problem in reduced coordinates, and an HPROM additionally hyperreduces its projected operators.

To express this common equilibrium problem in the coordinates used by the reduced models, we distinguish the physical nodal displacement vector $\mathbf{d} \in \mathbb{R}^{N_{\text{dof}}}$ from the independent state $\mathbf{u} \in \mathbb{R}^N$ obtained after enforcing periodicity and removing rigid translation. In the deployed convention,

$$\mathbf{d}(\mathbf{u}, \mathbf{e}) = \mathbf{T}\mathbf{u} + \mathbf{g}_{\text{per}}(\mathbf{e}), \quad \mathbf{R}(\mathbf{u}; \mathbf{e}) = \mathbf{T}^T \mathbf{f}_{\text{int}}(\mathbf{d}(\mathbf{u}, \mathbf{e})) = \mathbf{0}. \quad (44)$$

Here $\mathbf{T}$ identifies periodic degrees of freedom, and the lifting $\mathbf{g}_{\text{per}}$ supplies the strain-dependent displacement jumps and anchor values defined in Section 2. Thus any independent state $\mathbf{u}$ produces a physical field satisfying the prescribed kinematic constraints. If $\mathbf{K}_{\text{phys}} = \partial_{\mathbf{d}} \mathbf{f}_{\text{int}}$ is the assembled physical stiffness, the constrained HDM tangent is $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{R} = \mathbf{T}^T \mathbf{K}_{\text{phys}} \mathbf{T}$. We use the internal-force sign convention for all residuals.

The reduced bases below are built from snapshots of this affine-retaining state $\mathbf{u}$, rather than from the fluctuation coordinates $\mathbf{a}$ of Eq. (8). If $\mathbf{u}_{\text{aff}}(\mathbf{e})$ denotes the affine displacement at the independent nodes, the two descriptions satisfy $\mathbf{a} = \mathbf{u} - \mathbf{u}_{\text{aff}}(\mathbf{e})$. This coordinate choice leaves the periodic boundary conditions unchanged and preserves load-correlated information used by the strain-informed identification below. Consequently, the reported POD dimension characterizes this representation and its truncation tolerance; it is not an intrinsic material dimension.

4.2 POD approximation and the conventional PROM

The first reduction targets the number of equilibrium unknowns. We collect HDM solutions $\mathbf{u}^s$ at sampled macroscopic strains $\mathbf{e}^s$, with $s = 1, \dots, N_s$, and center them about a reference state $\mathbf{u}_{\text{ref}}$. Their singular value decomposition (SVD) is

$$\mathbf{S}_{\text{snap}} = [\mathbf{u}^1 - \mathbf{u}_{\text{ref}}, \dots, \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}] = \mathbf{U}_S \mathbf{\Sigma}_S \mathbf{Y}_S^T. \quad (45)$$

Proper orthogonal decomposition (POD) retains the columns of $\mathbf{U}_S$ associated with the largest singular values. Collecting n_{tra} of these modes in the orthonormal reduced-order basis (ROB) $\mathbf{V}_{\text{tra}}$ gives the conventional affine approximation

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}, \quad \mathbf{q}_{\text{tra}} \in \mathbb{R}^{n_{\text{tra}}}, \quad \mathbf{V}_{\text{tra}}^T \mathbf{V}_{\text{tra}} = \mathbf{I}. \quad (46)$$

The PROM therefore replaces the N independent displacement coordinates by $n_{\text{tra}} \ll N$ modal amplitudes. Here *linear PROM* describes only the affine approximation space; the constitutive law and the reduced equilibrium equations remain nonlinear.

The retained dimension is selected from the relative squared reconstruction error. For a Euclidean POD,

$$\epsilon_{\text{POD}}^2(n_{\text{tra}}) = \frac{\sum_{j > n_{\text{tra}}} \sigma_j^2}{\sum_j \sigma_j^2}. \quad (47)$$

This criterion measures how accurately the supplied displacement snapshots are compressed; it is not an error bound for stresses at unsampled strains. Force evaluation, constitutive differentiation, and equilibrium conditioning can make small displacement components important.

Once the approximation space is fixed, equilibrium is enforced by Galerkin projection. At a prescribed strain $\mathbf{e}$, the admissible variations are $\delta \tilde{\mathbf{u}} = \mathbf{V}_{\text{tra}} \delta \mathbf{q}_{\text{tra}}$; requiring zero residual work against them gives the reduced equations and tangent

$$\mathbf{r}_{\text{tra}}(\mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{V}_{\text{tra}}^T \mathbf{R}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{0}, \quad \mathbf{K}_{\text{tra}} = \mathbf{V}_{\text{tra}}^T \mathbf{K} \mathbf{V}_{\text{tra}}. \quad (48)$$

POD has reduced the nonlinear system to n_{tra} unknowns, but evaluating $\mathbf{R}$ and $\mathbf{K}$ still visits every microscopic element. Hyperreduction, introduced below, addresses this remaining cost.

4.3 Nonlinear reduction through latent-space closure

The POD dimension required to reconstruct a displacement field can exceed the number of coordinates needed to parametrize the loading family. Latent-space closure exploits this distinction: it retains a rich linear span, but learns dependencies among its modal amplitudes so that only a smaller subset remains unknown in equilibrium [22, 24]. We partition an $n_{\text{tot}} = n + \bar{n}$ dimensional ROB into orthonormal primary and secondary blocks,

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{V}^T \mathbf{V} = \mathbf{I}, \quad \bar{\mathbf{V}}^T \bar{\mathbf{V}} = \mathbf{I}, \quad \mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0}, \quad (49)$$

and project each snapshot onto the corresponding coordinates,

$$\mathbf{q}^s = \mathbf{V}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}), \quad \bar{\mathbf{q}}^s = \bar{\mathbf{V}}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}). \quad (50)$$

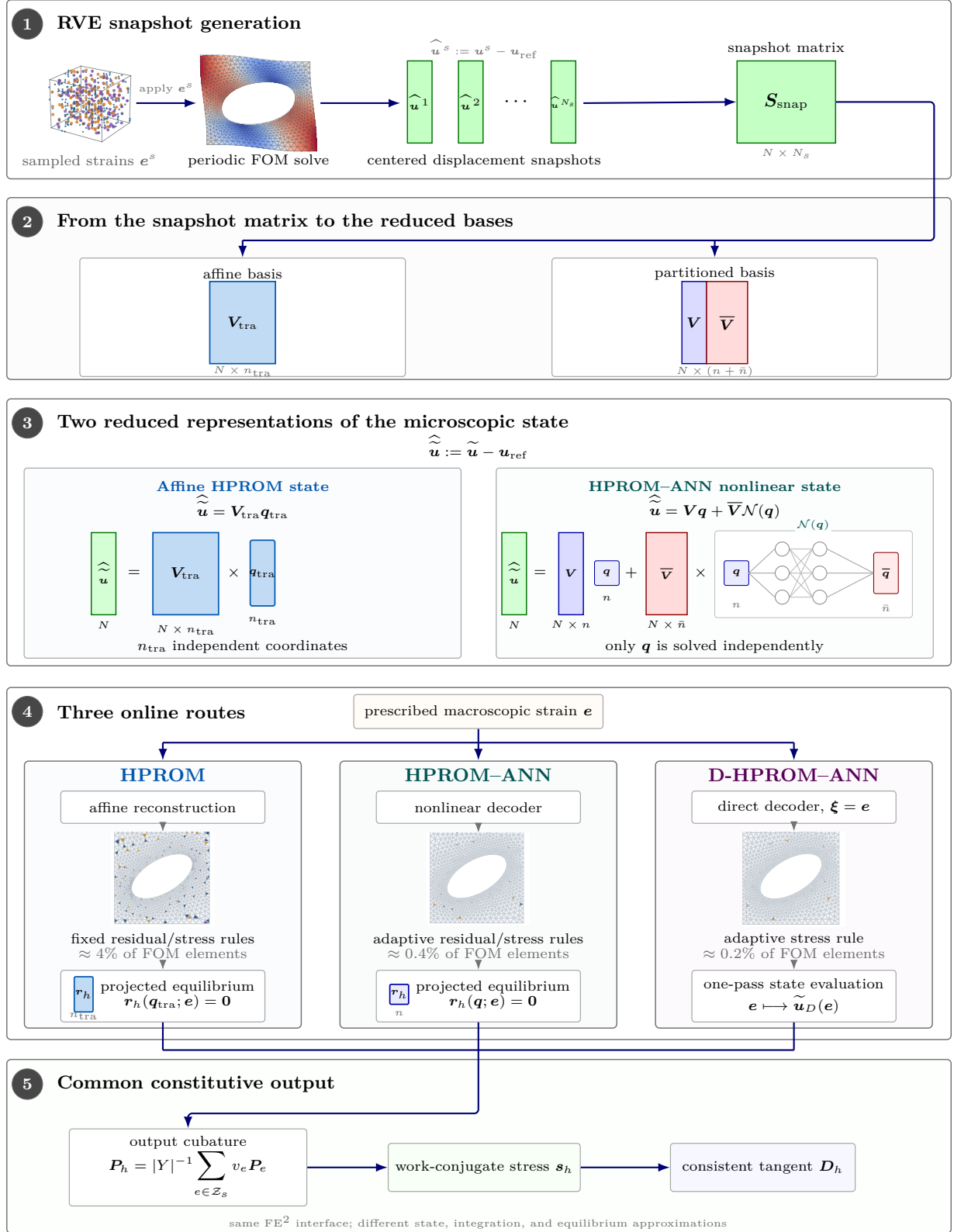


Figure 4: Intrusive reduced references used in the structural comparison. HDM snapshots provide either an affine state representation or a primary–secondary partition in which the ANN predicts the secondary coordinates. Online, HPROM and HPROM–ANN solve projected equilibrium with fixed and adaptive cubature, respectively, whereas D-HPROM–ANN evaluates the nonlinear decoder directly. All three retain microscopic constitutive evaluations and return the same stress–tangent interface.

The primary coordinates $\mathbf{q} \in \mathbb{R}^n$ remain independent unknowns, whereas an ANN closure $\mathcal{N} : \mathbb{R}^n \rightarrow \mathbb{R}^{\bar{n}}$ predicts the secondary coordinates. The resulting nonlinear decoder and its tangent are

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q}; \boldsymbol{\eta}), \quad \mathbf{B}(\mathbf{q}) = \partial_{\mathbf{q}}\tilde{\mathbf{u}} = \mathbf{V} + \bar{\mathbf{V}}\mathcal{N}_{,\mathbf{q}}. \quad (51)$$

Thus the ANN predicts neither stress nor energy: it predicts the amplitudes needed to reconstruct the microscopic displacement. The secondary modes remain in that field, but no longer appear as independent equilibrium unknowns.

The closure is trained on snapshot pairs $(\mathbf{q}^s, \bar{\mathbf{q}}^s)$ by minimizing

$$\mathcal{L}_{\text{cl}}(\boldsymbol{\eta}) = \frac{1}{|\mathcal{I}_{\text{fit}}|\bar{n}} \sum_{s \in \mathcal{I}_{\text{fit}}} \|\mathcal{N}(\mathbf{q}^s; \boldsymbol{\eta}) - \bar{\mathbf{q}}^s\|_2^2. \quad (52)$$

Input standardization is absorbed into the closure map, while the secondary outputs remain unscaled. Smooth tanh activations provide the decoder derivatives required by projection. This coordinate loss measures displacement reconstruction within the secondary space; it is not a stress- or energy-error guarantee.

The leading POD coefficients need not provide a well-conditioned primary-to-secondary map. We therefore use the input-informed construction of Ref. [39] to rotate the retained POD span and obtain three primary coordinates $\boldsymbol{\xi}$ approximately aligned with the three macroscopic strain components. If $\mathbf{A}_m$ denotes the resulting nonsingular coordinate transformation, the deployed decoder is

$$\tilde{\mathbf{u}}(\boldsymbol{\xi}) = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{A}_m\boldsymbol{\xi} + \bar{\mathbf{V}}\hat{\mathcal{N}}(\boldsymbol{\xi}), \quad \mathbf{B}_{\boldsymbol{\xi}} = \partial_{\boldsymbol{\xi}}\tilde{\mathbf{u}}. \quad (53)$$

The alignment $\boldsymbol{\xi}^s \approx \mathbf{e}^s$ is fitted from the snapshots; it is not a kinematic identity. The iterative PROM-ANN therefore solves at fixed $\mathbf{e}$ for the state satisfying

$$\mathbf{r}(\boldsymbol{\xi}; \mathbf{e}) = \mathbf{B}_{\boldsymbol{\xi}}^T \mathbf{R}(\tilde{\mathbf{u}}(\boldsymbol{\xi}); \mathbf{e}) = \mathbf{0}, \quad (54)$$

rather than imposing $\boldsymbol{\xi} = \mathbf{e}$. Because $\mathbf{B}_{\boldsymbol{\xi}}$ depends on the state, the exact Newton derivative includes the curvature of the decoder manifold. Appendix D gives the coordinate construction, the reconstruction-error split, and this tangent explicitly.

For the comparison below, the affine and nonlinear models share the same retained span, $n_{\text{tot}} = n_{\text{tra}}$. The nonlinear model reduces the number of equilibrium unknowns from n_{tra} to $n = 3$; hyperreduction has not yet been introduced.

For compactness, the cubature and output formulas below retain $\mathbf{q}$ for the primary modal amplitudes; in the deployed nonlinear model, the decoder and adaptive-weight maps are evaluated through $\boldsymbol{\xi} = \mathbf{A}_m^{-1}\mathbf{q}$.

4.4 Hyperreduction of the projected equations

Although the PROMs reduce the number of equilibrium unknowns, their projected equations still require full-mesh element evaluations. Hyperreduction addresses this remaining cost by replacing each full sum with a sparse weighted sum. It is therefore a second approximation, introduced after the displacement representation has been chosen.

Two integration tasks must be distinguished: assembling the projected equilibrium equations and recovering the homogenized stress. Hernández et al. [39] formulate both tasks and discuss the use of either a common or a dedicated output rule. We use separate elementwise rules because their integrands and accuracy requirements differ. The construction is summarized here; Appendix E records the matrix conventions and fitting losses used in our implementation.

4.4.1 Fixed elementwise cubature

Let $\mathcal{E}$ be the microscopic element set and $\mathbf{L}_e$ extract element nodal values from $\mathbf{d}$. At fixed macroscopic input, define the reconstructed element state and its reduced tangent by

$$\mathbf{d}_e = \mathbf{L}_e(\mathbf{T}\tilde{\mathbf{u}} + \mathbf{g}_{\text{per}}), \quad \mathbf{B}_e = \mathbf{L}_e\mathbf{T}\mathbf{B}. \quad (55)$$

The full reduced residual is $\sum_{e \in \mathcal{E}} \mathbf{h}_e$, where $\mathbf{h}_e = \mathbf{B}_e^T \mathbf{f}_{\text{int},e}$. Fixed ECM approximates it as

$$\mathbf{r}_{\text{ECM}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e \mathbf{h}_e(\mathbf{q}; \mathbf{e}) = \mathbf{0}, \quad \mathcal{Z}_r \subset \mathcal{E}, \quad w_e \geq 0. \quad (56)$$

For the conventional HPROM, $\mathbf{B}$ is replaced by $\mathbf{V}_{\text{tra}}$ and the unknown is $\mathbf{q}_{\text{tra}}$. There is no ANN in this tier. A selected element retains its ordinary within-element quadrature. Consequently, selected elements, selected Gauss points, and independent coordinates are distinct counts.

Offline, the element contributions are collected over the training states and compressed by SVD. ECM then selects a sparse support and nonnegative multipliers that reproduce the retained integration modes [34]. If $\mathbf{G}_{\text{cub}}$ denotes those modes augmented by the constant-integration constraint and $\mathbf{b}_{\text{cub}} = \mathbf{G}_{\text{cub}} \mathbf{1}$, selection stops when

$$\frac{\|\mathbf{G}_{\text{cub}} \mathbf{w} - \mathbf{b}_{\text{cub}}\|_2}{\|\mathbf{b}_{\text{cub}}\|_2} \leq \varepsilon_{\text{ECM}}. \quad (57)$$

The tolerance ε_{SVD} controls the preceding integrand compression, whereas ε_{ECM} controls the sparse fit. For the residual rule, the element data are $\mathbf{h}_e$. For the output rule, they are the four components of the element contribution to the average first Piola stress; all four are retained because this tensor is not generally symmetric. Both rules impose $\sum_e w_e = N_{\text{el}}$, with physical quadrature measures already included in the element contributions. Their supports and weights remain fixed online.

4.4.2 Manifold-adaptive weights

A fixed rule must represent every sampled state with the same multipliers. MAW-ECM instead retains one common support but lets its weights vary over the reduced manifold [39]. Applied to the residual, this gives

$$\mathbf{r}_{\text{MAW}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e(\mathbf{q}) \mathbf{B}_e(\mathbf{q})^T \mathbf{f}_{\text{int},e}(\mathbf{d}_e) = \mathbf{0}. \quad (58)$$

Starting from a fixed ECM rule, the MAW-ECM pruning algorithm removes elements while redistributing nonnegative statewise weights; its local and graph-regularized stages are described in Ref. [39]. Pruning identifies the support but not the continuous online map. We instantiate that final regression stage with neural fields. Because our equilibrium and stress rules are separate, we train two maps, $\mathcal{N}_r$ and $\mathcal{N}_s$, on independently pruned supports $\mathcal{Z}_r$ and $\mathcal{Z}_s$:

$$\begin{aligned} \mathbf{w}(\boldsymbol{\xi}) &= N_{\text{el}} \text{softmax}(\mathcal{N}_r(\boldsymbol{\xi}; \boldsymbol{\theta}_r)), & \sum_{e \in \mathcal{Z}_r} w_e &= N_{\text{el}}, \\ \mathbf{v}(\boldsymbol{\xi}) &= N_{\text{el}} \text{softmax}(\mathcal{N}_s(\boldsymbol{\xi}; \boldsymbol{\theta}_s)), & \sum_{e \in \mathcal{Z}_s} v_e &= N_{\text{el}}. \end{aligned} \quad (59)$$

Input standardization is absorbed into each network. The closure ANN $\mathcal{N}$ remains a different object: it predicts secondary displacement coordinates, not weights. The two weight fields are fitted against full-mesh element sums at each decoded training state. The scaled softmax guarantees positivity and normalization, but integration accuracy is learned. Thus the offline order is: train the closure, select and prune the two supports, and finally fit the two weight fields.

Online, the current reduced iterate is passed to both the closure and $\mathcal{N}_r$; $\mathcal{N}_s$ supplies the output weights. The supports and all network parameters remain fixed. Differentiating the adaptive residual produces three contributions: element stiffness, decoder curvature, and weight variation. We retain only the first in the microscopic modified-Newton iteration, while the consistent constitutive tangent differentiates the complete implemented residual and output, as described next. For the affine HPRM, the decoder curvature and weight-variation terms vanish, so its projected stiffness is the exact residual Jacobian.

4.5 Constitutive output

4.5.1 Equilibrium-based stress and consistent tangent

The reduced microscopic solver must return the same work-conjugate stress and tangent required by the FE² problem. The final stage of Figure 4 summarizes this common output path. Residual cubature alone is insufficient because equilibrium and effective stress involve different integrands. The separate output rule introduced above reconstructs the average first Piola stress as

$$\mathbf{P}_h(\mathbf{q}, \mathbf{e}) = \frac{1}{|Y|} \sum_{e \in \mathcal{Z}_s} v_e(\mathbf{q}) \sum_{g \in e} \omega_g \mathbf{P}_{\mu,g}(\mathbf{d}_e), \quad \mathbf{S}_h = \mathbf{F}^{-1} \mathbf{P}_h. \quad (60)$$

Here v_e is fixed for HPRM and state dependent for HPRM-ANN. Because the independently fitted output rule need not produce an exactly symmetric converted tensor, the implementation returns its symmetric work-conjugate components,

$$\mathbf{s}_h(\mathbf{q}, \mathbf{e}) = ((S_h)_{11}, (S_h)_{22}, \frac{1}{2}[(S_h)_{12} + (S_h)_{21}])^T, \quad (61)$$

and differentiates this actual output. The residual and stress supports may overlap; their union defines the element data required by the iterative constitutive evaluation, whereas only $\mathcal{Z}_r$ enters the equilibrium solve.

Reduced equilibrium defines an implicit state $\mathbf{q}^*(\mathbf{e})$ and hence the constitutive map $\mathbf{s}_h(\mathbf{q}^*(\mathbf{e}), \mathbf{e})$. If the residual Jacobian is nonsingular, differentiating equilibrium gives

$$\mathbf{D}_h = \mathbf{s}_{h,e} - \mathbf{s}_{h,q} \mathbf{r}_{h,q}^{-1} \mathbf{r}_{h,e}. \quad (62)$$

The partial derivatives hold the other argument fixed and are evaluated at the converged state. They include the periodic lifting, stress conversion, decoder curvature, and—for adaptive rules—weight variation. Equation (62), rather than the modified-Newton matrix used to reach microscopic equilibrium, supplies the tangent returned to the structural solver. Appendix E records how these derivatives are evaluated in the two implementations.

4.5.2 Direct decoder evaluation: D-HPROM-ANN

When input-informed coordinates closely approximate the loading parameters, the decoder can also be evaluated without solving reduced equilibrium, as discussed in the hyperelastic example of Ref. [39]. Here this defines a separate operating mode, D-HPROM-ANN, obtained by prescribing $\boldsymbol{\xi} = \mathbf{e}$ in Eq. (53):

$$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\mathbf{e}). \quad (63)$$

The decoded state is inserted directly into the adaptive stress rule, using the same closure and stress-weight field as the iterative model. Only the stress support is needed; no residual cubature or reduced equilibrium Jacobian is evaluated. The tangent is obtained by central differences of the complete direct stress map, updating the decoder, weights, lifting, and stress conversion at each perturbed input.

This route still evaluates the microscopic material law on selected elements; it is not a neural stress regressor or a learned macroscopic potential. The prefix D is our label for the direct operating mode, not a new projection method. Its accuracy depends on both the closure and the approximation $\boldsymbol{\xi} \approx \mathbf{e}$. Conversely, solving projected equilibrium corrects the primary state only within the chosen manifold and cubature approximation, so it does not guarantee a smaller stress error than direct evaluation at every input.

4.6 Mechanical structure retained by the reduced models

Unlike a direct constitutive surrogate, every reduced model above continues to evaluate the original microscopic hyperelastic law; the iterative variants also enforce projected equilibrium. This preserves a direct micromechanical link, but it does not automatically make the deployed macroscopic stress the derivative of an effective potential.

For fixed weights and consistent kinematics, Galerkin internal forces can be derived from a weighted reduced energy, as in the structure-preservation results established for ECSW under their stated hypotheses [33]. Two features of the present deployment prevent us from transferring that conclusion without further assumptions. First, equilibrium and stress use independently fitted cubature rules and therefore need not be derivatives of the same energy. Second, when the weights depend on the reduced state, weighting energy derivatives is not equivalent to differentiating the weighted energy:

$$\nabla_{\mathbf{q}} \sum_e w_e(\mathbf{q}) \Pi_e(\mathbf{q}, \mathbf{e}) = \sum_e w_e(\mathbf{q}) \nabla_{\mathbf{q}} \Pi_e + \sum_e \Pi_e \nabla_{\mathbf{q}} w_e. \quad (64)$$

The final term is absent from Eq. (58), and nonnegative weights do not eliminate it. Consequently, we claim neither an exact effective potential nor a polyconvexity certificate for the deployed HPROM variants. Their consistent tangent is the derivative of the implemented stress map, not a certificate of conservativity or tangent symmetry.

4.7 Offline preparation and online evaluation

The accuracy–cost comparison in Section 5.3.5 concerns online constitutive evaluation. Before deployment, the offline stage collects HDM snapshots, constructs the reduced spaces and latent coordinates, fits the nonlinear closure, selects the cubature supports, trains adaptive weights where applicable, and builds the reduced meshes. These preparation costs are not included in the reported online timings.

To translate sparse cubature into wall-clock savings, each optimized reduced mesh contains only the selected elements and the nodes they require. The corresponding basis operators and periodic lifting are restricted to that support before online evaluation. Selected elements retain the original geometry, constituent

law, and quadrature; their cubature weights approximate the omitted contributions. At each online query, the model reconstructs the required displacement entries, solves reduced equilibrium when its operating mode requires it, and returns effective stress and tangent.

Online cost therefore depends on the support size, the retained secondary dimension, the equilibrium iterations, and the derivative evaluation—not only on the number of primary coordinates. Table 2 closes the construction by comparing the three deployed evaluators with their HDM/FOM reference.

Table 2: Intrusive evaluators used in the structural accuracy–cost comparison and their HDM/FOM reference. All reduced models retain microscopic constitutive evaluations.

Model	State representation	Equilibrium unknowns	Integration	Learned component
HDM / FOM	$\mathbf{u} \in \mathbb{R}^N$	N	Full mesh	None
HPROM	$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}$	n_{tra}	Fixed ECM	None
HPROM-ANN	$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q})$	n	Adaptive ECM	Secondary coordinates; residual and stress weights
D-HPROM-ANN	$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\mathbf{e})$	None	Adaptive stress rule	Same closure and stress weights as HPROM-ANN

5 Numerical examples

We use two periodic microstructures, each with its own independently trained models, so that the assessment does not rest on a single cavity arrangement. The multicavity RVE (MC-RVE) presents several geometric directions within one cell, which makes it a natural setting in which to isolate the effect of learning the paired features and of changing their number. The single-cavity RVE (SC-RVE) is instead the microscopic model subsequently propagated through the structural FE² calculation and shared with the intrusive references of Section 4; it therefore supplies the deployment qualification. Together, the two benchmarks test the same constitutive construction in distinct geometric settings, but not transfer of parameters between them.

The evidence follows those roles in causal order. After the common setting, we first ask on the MC-RVE how many paired features are needed and what learning them adds. We then ask on the SC-RVE whether the selected learned energies are accurate enough for the structural calculation of Section 5.3.5. Throughout, analytical guarantees are distinguished from predictive accuracy and finite implementation checks.

5.1 Common setting

Both benchmarks are square periodic cells with total void fraction 0.20. Their solid phase follows the same compressible Neo-Hookean law in plane strain. With $\mathbf{C}_\mu = \mathbf{F}_\mu^T \mathbf{F}_\mu$ and $J_\mu = \det \mathbf{F}_\mu > 0$, its local energy density is

$$\psi_\mu = \frac{\mu}{2}(\text{tr } \mathbf{C}_\mu - 2) - \mu \log J_\mu + \frac{\lambda}{2}(\log J_\mu)^2, \quad (65)$$

with Young’s modulus 1.628 GPa and Poisson ratio 0.4, where μ and λ are the Lamé constants. The cavity geometry, rather than the matrix law, distinguishes the two benchmarks.

A common diagnostic state. The two cells respond differently to the same multiaxial deformation. Figure 5 applies $(E_{11}, E_{22}, 2E_{12}) = (18, -4, 8)\%$ to both FOMs. The inclined cavity of the SC-RVE concentrates the response around one dominant geometric direction, whereas the nonaligned cavities of the MC-RVE create several interacting localization regions. This geometric contrast motivates the distinct roles of the two benchmarks; the figure shows FOM fields only and is not an error test.

5.1.1 Training protocol

The two benchmarks answer different questions, but their energies are trained and selected in the same way, so that differences within each benchmark reflect the constitutive representation rather than the fitting procedure. Every model is a scalar energy $W(\mathbf{e}; \boldsymbol{\theta})$, whose first derivative gives the stress and whose second

FOM microscopic fields at a common diagnostic state $(E_{11}, E_{22}, 2E_{12}) = (18, -4, 8)\%$

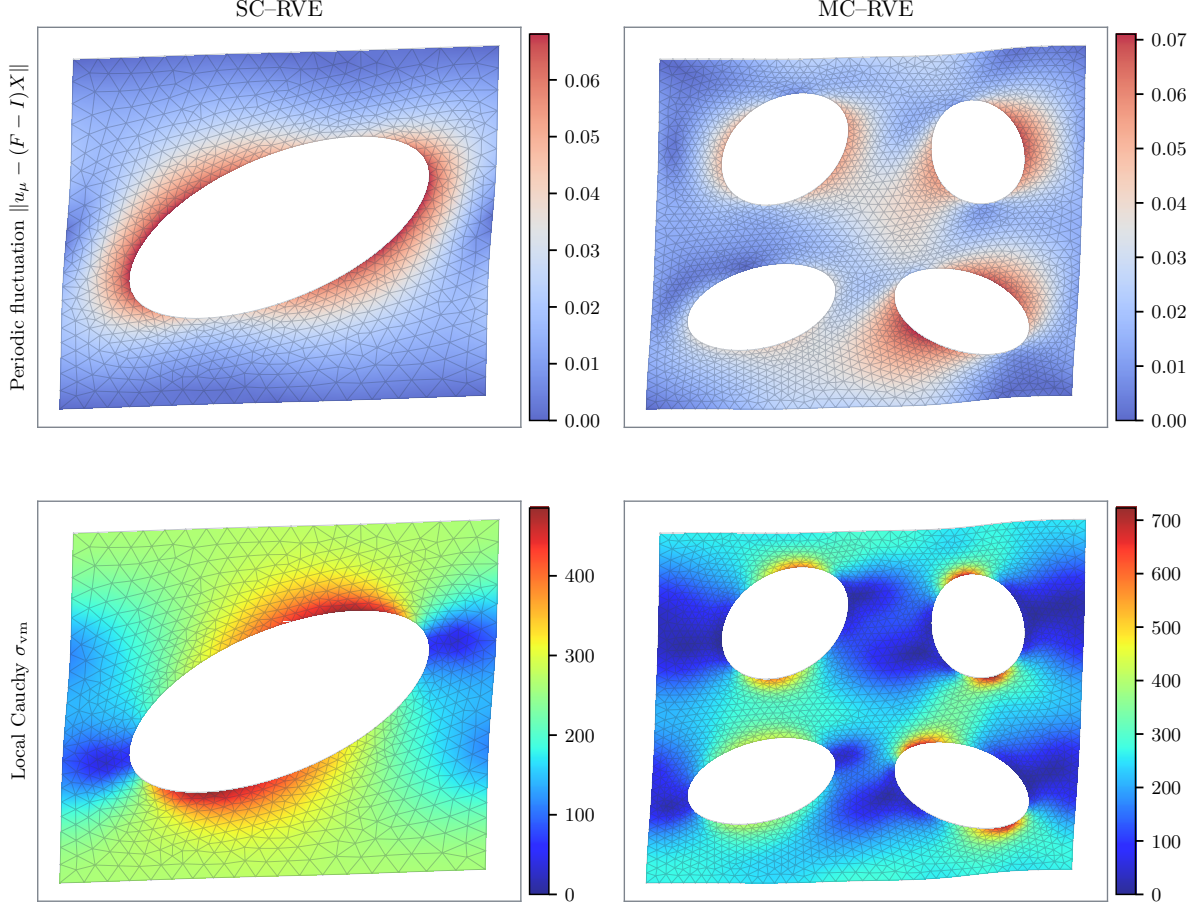


Figure 5: FOM microscopic fields at the common diagnostic state $(E_{11}, E_{22}, 2E_{12}) = (18, -4, 8)\%$, shown on the actual deformed cells without displacement amplification. Top: magnitude of the periodic displacement fluctuation after removing the imposed affine part. Bottom: local three-dimensional Cauchy von Mises stress, including the plane-strain out-of-plane stress; Gauss-point values are averaged to the nodes for visualization. Each panel uses its own color limits to resolve the spatial pattern within its respective cell; colors are therefore not a cross-cell magnitude comparison.

derivative gives the tangent $\mathbf{D}$ of Eq. (3). It is fitted to FOM energies and stresses at the fitting states and to the effective tangent $\mathbf{D}_0 = \mathbf{D}(\mathbf{0})$ of the periodic cell at the undeformed state.

The ICNN of Eq. (29) and the ICKAN of Eq. (30) have two hidden layers of widths 24 and 12, respectively, and each ICKAN connection combines six integrated-hat bases on fixed knots from -1.2 to 1.2 ($h = 0.48$). With six fixed paired features, the ICNN has 945 trainable parameters and the ICKAN 1629; learning the features adds only five parameters per feature. The initial features are selected from a finite bank of admissible candidates using only the fitting response and the reference tangent.⁴ Before entering either core, each feature is centered at its undeformed value $z_i^0 = 1/p_i + 1/q_i$ and divided by its largest deviation from that value over the fitting states at initialization. The conditioned inputs therefore start in $[-1, 1]$, which the ICKAN knots cover with a small margin. The Unconstrained energy uses three softplus hidden layers of widths 128, 128, and 64 on the inputs of Eq. (19), for 25 473 parameters.

All trainable parameters, including the feature parameters of the learned variants, minimize

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{\sum_k \|\mathbf{s}(e_k; \boldsymbol{\theta}) - \mathbf{s}_k\|_2^2}{\sum_k \|\mathbf{s}_k\|_2^2} + 0.2 \frac{\sum_k [W(e_k; \boldsymbol{\theta}) - W_k]^2}{\sum_k W_k^2} + 0.02 \frac{\|\mathbf{D}(\mathbf{0}; \boldsymbol{\theta}) - \mathbf{D}_0\|_F^2}{\|\mathbf{D}_0\|_F^2}, \quad (66)$$

⁴The bank contains 2028 candidates, which combine twelve orientations $\theta_i = k\pi/12$ with the stretch powers $p_i, q_i \in \{0.5, 1, 1.5, 2, 3\}$ and with determinant exponents at the bounds and midpoint of Eq. (26), moved slightly inside the admissible interval so that the parameterization of Eq. (27) can represent them. Nonnegative least-squares fits of the fitting response and of the reference tangent identify the contributing candidates. The largest contributions are retained when more than m are selected, and pivoted QR completes the set when fewer are.

where the sums run over the fitting states $\mathbf{e}_k$, with FOM energies W_k and stresses $\mathbf{s}_k$. Stress carries the largest weight because it is queried throughout the FE² calculation. The energy term keeps the potential itself, and not only its gradient, close to the homogenized energy. The small tangent term anchors the stiffness at the undeformed state, where the vanishing stresses contribute little to the first term.

Optimization is full-batch and in double precision. The validation error is the first term of Eq. (66) evaluated on the validation states. Adam runs until this error stops improving, or for at most 200 000 steps, and L-BFGS then refines the best Adam state.⁵ The retained checkpoint is the one with the lowest validation error, and each model is trained from three seeds, 16, 29, and 47. Independent test states are evaluated only after every checkpoint has been frozen.

5.2 Multicavity RVE: learning the paired features

The paired-feature construction guarantees admissibility for every allowed parameter set, but it does not determine how much directional information the convex core needs. This question is particularly relevant for the MC-RVE, whose nonaligned cavities introduce several geometric directions. A small prescribed feature set may then fail to distinguish deformation states with different directional responses. One remedy is to supply more features; another is to let their directions and exponents adapt to the RVE response. We examine both effects together by asking how accuracy changes with the number m of paired features and, for each m , what is gained by learning rather than fixing their parameters.

5.2.1 Cell, data, and controlled comparison

The MC-RVE distributes the void space among four unequal, off-center ellipses with different aspect ratios and orientations; see Fig. 6. Its several geometric directions make it the representation benchmark. The working mesh has 4621 quadratic triangles.

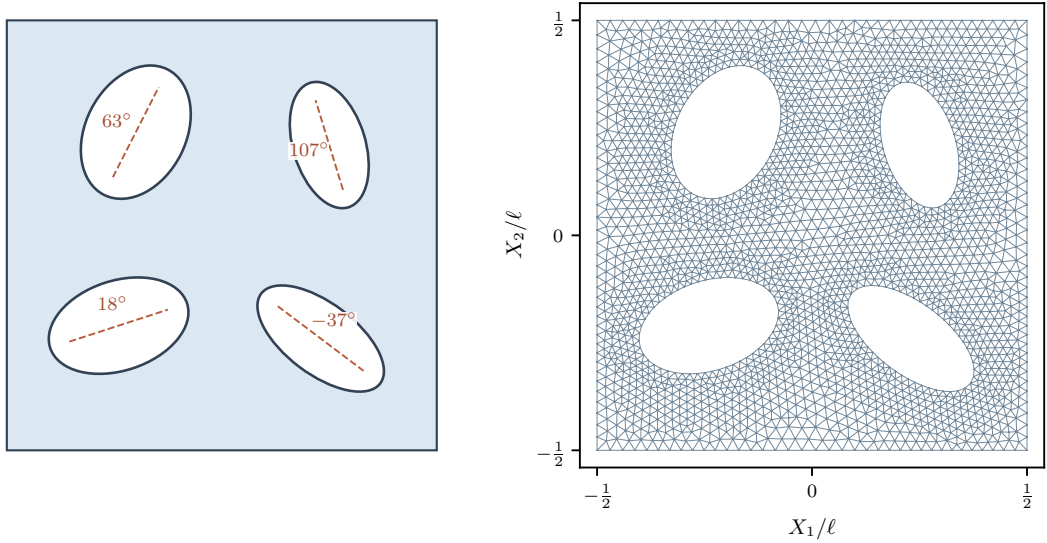


Figure 6: MC-RVE cavity geometry (left) and quadratic mesh (right). Dashed lines identify cavity major axes.

Refining the mesh to 8961 elements changes the homogenized energy, stress, and tangent by less than 0.1% over 64 audited states that include the corners of the strain domain used below; the sampled maximum microscopic first Piola stress norm changes by at most 4.06%. These comparisons support the working mesh for the representation study.

Controlled comparison. We sample $E_{11}, E_{22} \in [-0.04, 0.20]$ and $2E_{12} \in [-0.08, 0.08]$. The left panel of Fig. 7 shows the domain. The 4200 fitting states combine 4096 scrambled Sobol points in the interior with 96 points on the faces and the eight corners, so that the boundary of the box is sampled rather than reached

⁵Adam starts from learning rates of 5×10^{-3} for the ICNN and ICKAN and 4×10^{-4} for the Unconstrained energy, halved after 40 validation checks without improvement down to 10^{-5} , with a check every ten steps. It stops when the best validation error improves by less than 0.1% over 100 checks at the smallest learning rate. L-BFGS, in calls of at most ten iterations, stops under the same criterion over 30 calls.

by extrapolation. Independent Sobol sequences supply 512 validation and 512 independent-test states. Ten further straight paths run from the undeformed state to the boundary of the box in axial, biaxial, shear, and combined directions, each with 40 scored states; like the test states, they play no role in training or selection. For each $m \in \{2, 4, 6, 8, 16, 24, 32\}$, ICNN and ICKAN are trained in two variants that start from the same m initial features. The fixed variant holds their directions and exponents constant while training the convex core, whereas the learned variant optimizes those parameters subject to the constraints of Section 3. All other ingredients of Section 5.1.1 are shared within each core, so the difference between the two variants isolates the effect of adapting the features.

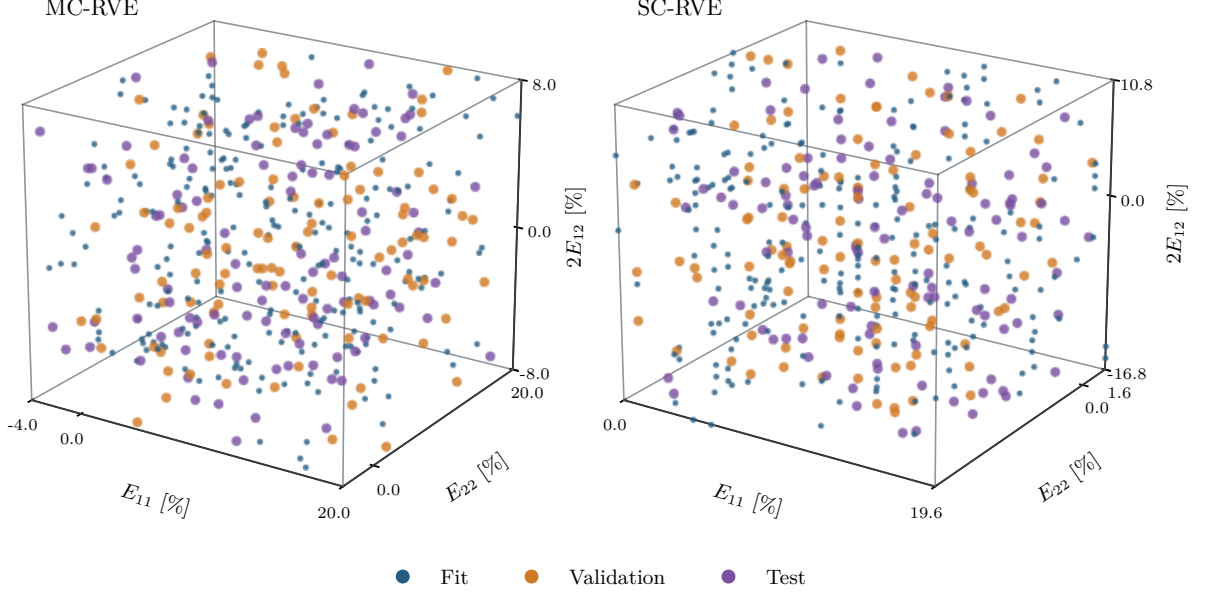


Figure 7: Constitutive sampling domains, in percent. Left: MC-RVE representation benchmark. Right: SC-RVE deployment benchmark. Wireframes show the prescribed limits; colors identify fitting, validation, and independent-test states.

5.2.2 How many paired features are needed?

Figure 8 answers this question in the same way for both convex cores. Two paired features leave the response under-resolved, and both variants improve as m increases, but in different ways. The learned variants gain most up to $m = 6$ and change by less than a factor of two beyond it, whereas the fixed variants continue to improve gradually up to $m = 32$. We therefore select $m = 6$, the first count at which both learned cores enter their low-error regime, for independent evaluation.

5.2.3 What does learning the features add?

Table 3 answers this question on the independent test states. At $m = 6$, learning the paired features lowers the ICNN median stress error from 0.328% to 0.0445% and the ICKAN error from 0.301% to 0.0406%, a reduction by a factor of approximately 7.4 in either core. The energy and the tangent, which is never fitted away from the undeformed state, improve by factors of about 14 and 5, respectively. Adding prescribed directions does not close this gap. In the validation sweep, even 32 fixed features, with more than twice the trainable parameters of the learned six-feature models, remain about three times less accurate in relative stress error, a factor of about ten in the squared validation error of Fig. 8. The ten held-out paths confirm the ordering for stress, energy, and tangent on every path. The largest gain appears in pure shear, where the relative stress error falls from 3.2–4.0% with fixed features to 0.47–0.58% with learned ones, whereas biaxial compression produces the largest tangent errors for both variants. The similar gain for ICNN and ICKAN shows that the central improvement comes from adapting the directional representation rather than from one particular convex core.

The Unconstrained energy places these values in context. Without convexity or growth conditions, it reaches 0.0073% on the same states, about six times below either learned constrained model. Its network is also considerably larger. At matched capacity, the controls of Appendix F place the in-domain cost of the

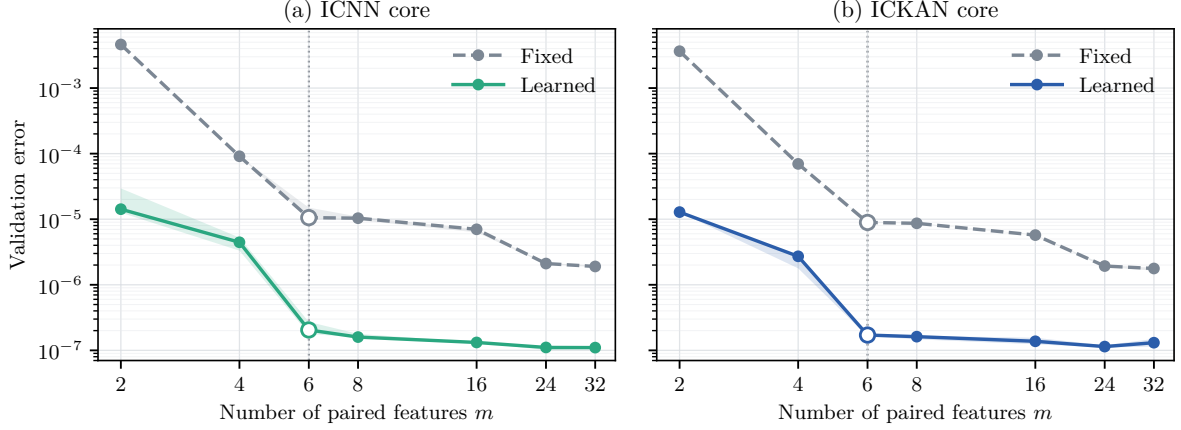


Figure 8: MC-RVE validation error, the relative squared stress error of Section 5.1.1, versus the number m of paired directional features. Lines show the median over three seeds and bands span their minimum–maximum range. The vertical guide marks $m = 6$, used for the independent comparison.

constraints on this cell at a factor of 3.2 to 5.5 in stress error; the one exception, a wide ICKAN, reflects early stopping rather than capacity.

Table 3: MC-RVE independent-test errors at $m = 6$. Each entry is the relative error norm of the stress $\mathbf{s}$, the energy W , or the tangent $\mathbf{D}$ of Eq. (3) over the 512 test states, as the median over three seeds. For every quantity, each learned seed is more accurate than each fixed seed. Adam stopped at a validation plateau for the fixed variants, whereas five of the six learned runs and the Unconstrained energy, which has no paired features, reached the 200 000-step cap.

Model	Stress [%]	Energy [%]	Tangent [%]
ICNN, fixed	0.3282	0.1096	1.9247
ICNN, learned	0.0445	0.0079	0.3851
ICKAN, fixed	0.3014	0.0943	1.7600
ICKAN, learned	0.0406	0.0066	0.3622
Unconstrained energy	0.0073	0.0011	0.0857

5.2.4 What do the constraints guarantee beyond the data?

For the ICNN and ICKAN energies, conditions (C1)–(C6) hold for every admissible parameter set by construction, whereas the Unconstrained energy is built to satisfy (C1)–(C4) only. Inside the box the difference does not show, since a bounded rank-one search over 520 states and 32 directions finds curvature above 236 MPa for all fifteen fits of Table 3. Farther out the two classes separate. Figure 9(a) follows equibiaxial compression, $\mathbf{F} = \sqrt{J} \mathbf{I}$, beyond the box. The minimum rank-one curvature of every Unconstrained fit changes sign between $J = 0.71$ and 0.73 , whereas the learned constrained fits remain above 220 MPa. A directed search over principal stretches between 0.55 and 1.5, in arbitrary orientations, shows that this loss is not tied to one path. It finds negative curvature for all three Unconstrained fits at more than a third of its 4096 sampled states, confirmed by energy finite differences, and only positive curvature for the twelve constrained fits, as their construction requires.

Figure 9(b) continues the same path toward volume collapse. The constrained energies diverge, as their logarithmic barrier guarantees, whereas the Unconstrained energy levels off at finite limits between 397 and 440 MPa. This contrast reflects the growth terms rather than polyconvexity, and an unconstrained network could receive the same barrier. No data constrain either energy at these states, so only the construction can guarantee (C5) and (C6) there.

Global nonnegative energy, which the construction leaves to the saved parameters, is certified by the sufficient bound of Appendix C for all six learned fits, with interval margins between 8.5 and 10.8 in normalized energy units. For all fifteen fits, the reference energy and stress vanish to round-off, and centered differences reproduce the stress and the tangent to better than 10^{-9} of their largest entries.

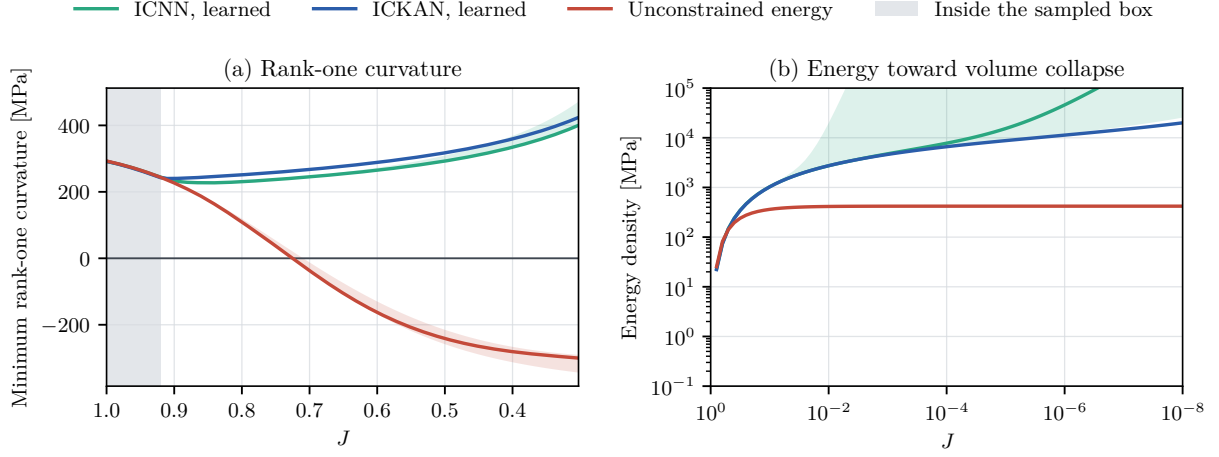


Figure 9: MC-RVE energies beyond the data along equibiaxial compression, $\mathbf{F} = \sqrt{J} \mathbf{I}$, for the learned ICNN and ICKAN energies and the Unconstrained energy. Lines show the median of three seeds and shaded bands their range. (a) Minimum rank-one curvature over unit directions. (b) Energy density toward volume collapse.

Taken together, the MC-RVE supports the choices carried into the SC-RVE study. Six paired features give a compact representation, and adapting their directional parameters supplies the main accuracy gain for either convex core. The learned constrained energies then keep the median stress error below 0.05% while securing, beyond the data, the rank-one convexity and growth that the Unconstrained energy loses there.

5.3 Single-cavity RVE: constitutive accuracy and FE² deployment

5.3.1 Cell and coupon-derived data

The SC-RVE contains one centered elliptical void with aspect ratio 2 and major-axis angle 30° relative to the cell axes; see Fig. 10. Its working mesh has 1546 quadratic triangles and 6320 independent microscopic displacement coordinates.

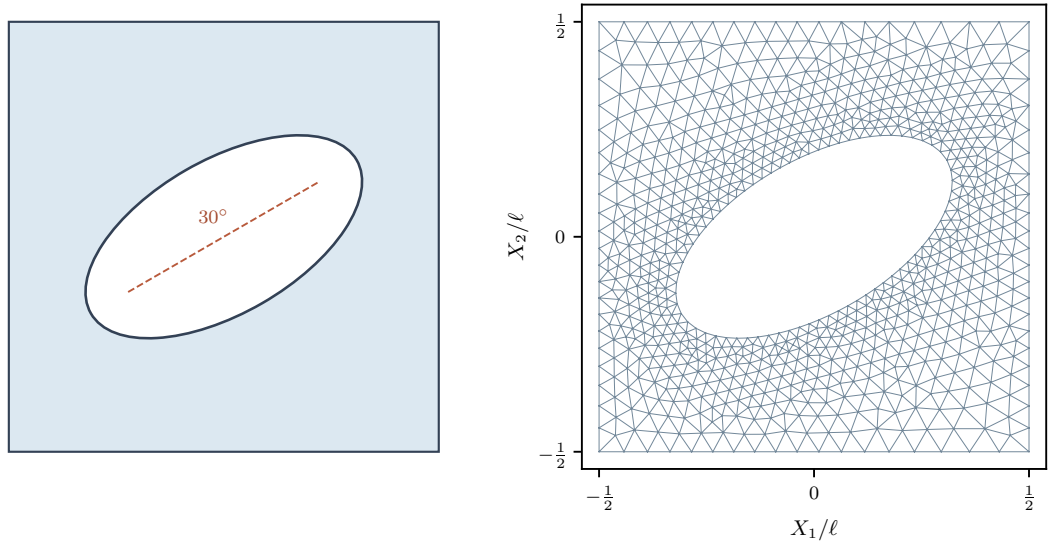


Figure 10: SC-RVE cavity geometry (left) and quadratic mesh (right). The dashed line identifies the cavity major axis.

At a uniaxial-stress state with $E_{11} = 0.20$, refining the mesh to 3095 elements changes the maximum microscopic first Piola stress norm by approximately 0.486% and the average axial stress by approximately 0.001%. This state lies on the path followed by the narrow section of the coupon in Section 5.3.5, beyond the largest axial strain reached there, so the check supports the mesh used for deployment.

Coupon-derived data. Unlike the MC-RVE, whose box is prescribed, the SC-RVE must supply the learned constitutive response throughout the coupon FE^2 calculation of Section 5.3.5, whose macroscopic quadrature points visit different strains during loading. We first ran a displacement-controlled coupon prepass with a St Venant–Kirchhoff approximation based on the cell reference tangent. The recorded strain envelope guides sampling only; RVE stresses provide the training labels. A second prepass with a softer response shifted the componentwise extrema by at most 5.1%, so we enlarged the spans by 40%. Because the coupon is tensile, the lower E_{11} bound is zero. In the engineering-strain coordinates $\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T$, the rounded bounds are

$$\mathbf{e}_{\min} = (0, -0.099, -0.168)^T, \quad \mathbf{e}_{\max} = (0.196, 0.016, 0.108)^T. \quad (67)$$

A Cartesian grid supplies 4950 nonreference states, of which 4208 are used for fitting and 742 for validation (split seed 5); an independent off-grid set of 400 states tests interpolation; see the right panel of Fig. 7. The same grid also provides neighboring states for the reduced microscopic models of Section 4.

5.3.2 Constitutive accuracy

The MC-RVE study identifies $m = 6$ as the first compact count at which both learned cores enter their low-error regime. We therefore ask whether that same count is sufficient for the SC-RVE deployment, without transferring feature parameters between the two geometries. Six initial features are selected anew from the SC-RVE fitting response and reference tangent, and the learned ICNN and ICKAN are trained as described in Section 5.1.1, as is the Unconstrained energy. For each model, the seed with the lowest validation error supplies the deployed model. Both constrained selections fall on seed 16, with the validation errors of the three seeds within 15% of one another, and the Unconstrained selection falls on seed 47.

Independent response. The 400 off-grid states now answer the deployment question directly. The compact ICNN and ICKAN attain aggregate stress errors of 0.621% and 0.636%, respectively, and the Unconstrained energy reaches 0.0052%, about 120 times lower; see Table 4. On this cell the construction therefore costs far more accuracy than on the MC-RVE, and capacity is not the reason. In the controls of Appendix F, cores widened to about 25 000 parameters change the constrained errors by less than 3%, whereas an Unconstrained energy with only 981 parameters remains about 75 to 80 times more accurate than either wide core. Thirty-two learned features lower the errors only to 0.537% and 0.618%. This behavior is consistent with the ordering argument of Section 3.3, since a monotone core cannot reverse the componentwise ordering of its features whatever its width; widening the core therefore cannot enlarge the response family beyond what the features and that ordering admit. The price of this restriction depends on the cell, a factor of 3.2 to 5.5 on the MC-RVE and of 76 to 126 on the SC-RVE at matched capacity. Comparable costs are reported for other constrained constructions. Linden et al. [14] found that imposing all common conditions on a small invariant-based network raised its stress error by about one order of magnitude for transverse isotropy, and still recommended the constrained model for its extrapolation. On homogenization data of a matrix with a stiff spherical inclusion, the polyconvex models of Klein et al. [44] reached root-mean-square stress errors three to five times those of their non-polyconvex counterparts, and Kalina et al. [43] dropped polyconvexity after finding it too restrictive for their fits. The MC-RVE factors are of this size, whereas the SC-RVE shows that the price can be much larger for some microstructures.

Response beyond the box. A separate 350-state probe resolves the response outside the fitting box. It contains five increasing overshoot factors and labelled face, edge, and corner patterns; five states without a finite FOM label are excluded for every model, leaving 345. The aggregate probe errors rise to 10.6% and 10.1% for ICNN and ICKAN and to 0.668% for the Unconstrained energy, with the ringwise results in Supplementary Section S1. On this cell the constraints thus cost accuracy inside and outside the box. Inside it, where every coupon state lies, both constrained energies remain below 0.65%; what the constraints secure beyond it is examined next. Thirty-two features would also raise the probe errors to 35.6% and 16.9% and lose the nonnegative-energy certificate of Appendix C, so the coupon keeps the $m = 6$ models.

Table 4: SC-RVE independent-test constitutive errors of the selected models, in percent.

Model	e_S^{test} [%]	e_W^{test} [%]	Worst test [%]
ICNN	0.6214	0.1929	4.9263
ICKAN	0.6359	0.1900	4.8240
Unconstrained energy	0.0052	0.0010	0.0203

All values are percentages. Stress and energy errors are aggregate array norms; the last column is the maximum per-state stress error. SC-RVE: 400 independent test states.

5.3.3 Admissibility beyond the data

The comparison of Section 5.2.4 carries over to this cell; Supplementary Section S2 gives the sampled values. On the test, probe, and 12 000 additional in-box states, the rank-one curvature of all three energies stays above 135 MPa. Farther out, a radial expansion of the box to four times its size finds negative curvature for the Unconstrained energy at 14 108 of 109 743 states, first at 1.55 times the box and $J = 0.78$, and a broad scan of principal stretches between 0.25 and 3 finds it at 10 048 of 24 000 states. The ICNN and ICKAN remain above 176 MPa on both sets. The Unconstrained energy also stays finite along $\mathbf{F} = \sqrt{J} \mathbf{I}$ as $J \rightarrow 0^+$, because its network inputs remain bounded on that path, whereas the logarithmic barrier makes the constrained energies diverge. For the two deployed models, the sufficient bound of Appendix C certifies nonnegative energy, with interval margins of 4.85 and 4.91 in normalized energy units. For all three energies, derivative checks and the vanishing closed-cycle work of Supplementary Section S3 confirm the potential structure of the implementation.

5.3.4 Reduced models built from the same snapshots

The 4950 displacement snapshots provide the basis of the reduced microscopic models as well as the effective constitutive labels. Figure 11 shows how the relative truncation norm of their POD basis decays with the number of modes; for the retained 39-dimensional space it is 9.94×10^{-7} . After its orthogonal rotation into primary and secondary spaces, $n = 3$ and $\bar{n} = 36$ in the notation of Section 4.

The fitted relation between the deployed primary coordinates and macroscopic strain has an aggregate relative array discrepancy of 3.60×10^{-5} on these snapshots; its maximum absolute component discrepancy is 2.55×10^{-5} . These quantities explain why macroscopic strain is a useful primary-coordinate predictor for this material. The direct substitution in D-HPROM-ANN nevertheless remains an additional approximation.

The secondary-coordinate closure is a 3–64–64–36 feed-forward network with two tanh hidden layers and a linear output. Its training script uses input standardization, the raw secondary-coordinate loss in Eq. (52), and a 90/10 internal fit/validation split with seed 20260903. Full-batch Adam has initial learning rate 2×10^{-3} , a validation-driven reduction schedule, and a maximum budget of 20 000 epochs. Its best relative validation mean-square error is 3.75×10^{-7} , an internal closure metric; the coupon of Section 5.3.5 provides the independent assessment.

Each deployed adaptive cubature field has architecture 3–128–128–128–10, GELU hidden activations, and the scaled softmax output of Eq. (59). The residual and stress checkpoints use distinct ten-element supports. Their larger neural cores illustrate why the online cost cannot be inferred solely from the three equilibrium coordinates or from the element count.

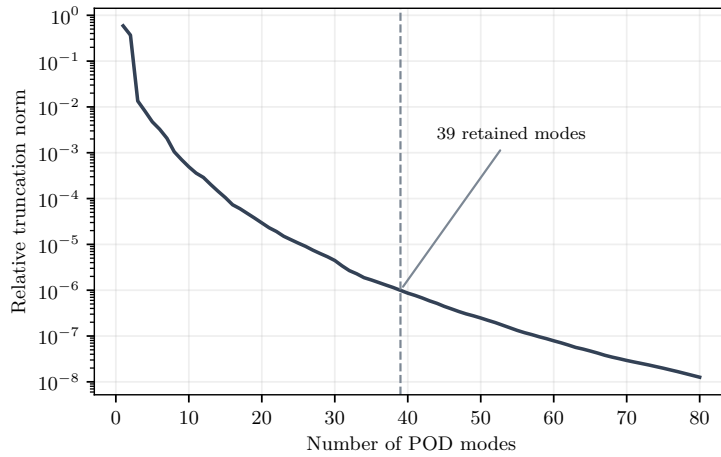


Figure 11: Relative truncation norm of the POD basis of the 4950 SC-RVE displacement snapshots versus the number of modes. The deployed basis retains 39 modes; its rotation into primary and secondary coordinates leaves this span unchanged.

Online models. The linear HPROM uses $n_{\text{tra}} = 39$ generalized coordinates, a residual rule supported on 135 elements, and a stress rule on 73 elements; their union contains 183 elements. The integrand-SVD tolerances are $\varepsilon_{\text{SVD}}^r = 10^{-3}$ and $\varepsilon_{\text{SVD}}^s = 10^{-5}$, retaining 134 and 72 integrand modes, respectively. The greedy integration tolerance in Eq. (57) is $\varepsilon_{\text{ECM}} = 10^{-6}$ for both rules, including the multiplier-sum constraint.

The iterative HPROM-ANN uses $n = 3$ primary and $\bar{n} = 36$ secondary coordinates, with ten residual and ten stress elements and a union of 19. Initial fixed-ECM candidates use $\varepsilon_{\text{SVD}} = 10^{-4}$ and $\varepsilon_{\text{ECM}} = 10^{-6}$; adaptive pruning targets ten elements per rule. This count is a prescribed pruning target, assessed through integration and response errors. The pruning uses up to 500 sampled states, tests up to 20 removal candidates per elimination, and applies only the local, positivity-enforcing stage of the algorithm; the residual weight field is then refitted on the same support with a longer training budget.

The direct model evaluates stress on ten elements and solves no microscopic equilibrium equations. All these counts refer to reduced computational meshes, not full-mesh evaluations followed by masking. Figure 12 shows their spatial distribution and the overlap of residual and output rules.

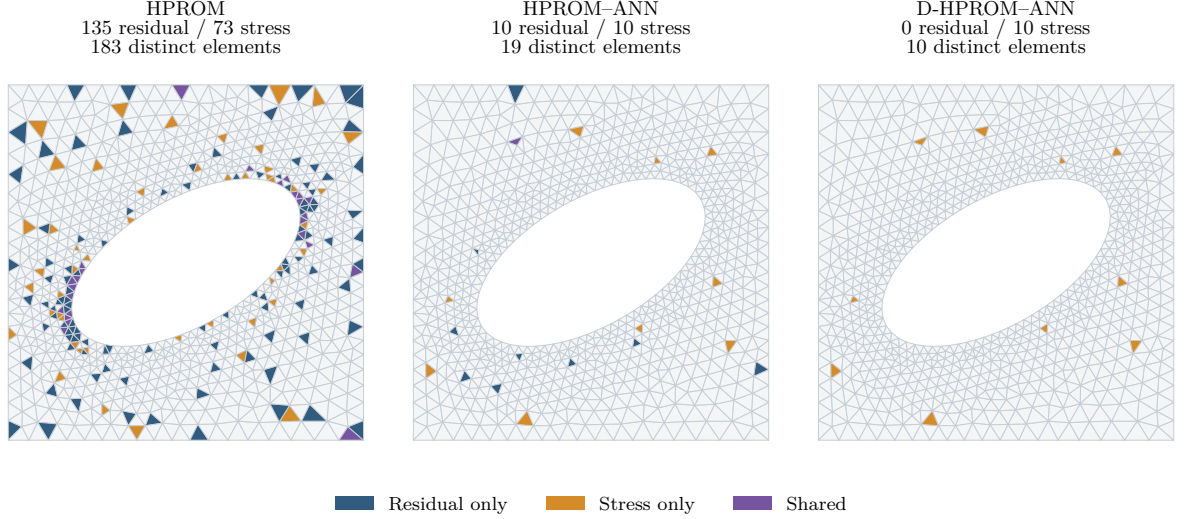


Figure 12: Element supports of the deployed hyperreduced microscopic models. The pale full-cell mesh provides location context only; discarded elements are not assembled online. Colors distinguish residual-only, stress-only, and shared elements.

5.3.5 Coupon FE^2 : accuracy and online cost

Purpose, boundary conditions, and coverage. We now use the SC-RVE to compare learned constitutive laws and reduced microscopic solvers within the same nonlinear structural calculation. The coupon measures how their approximations affect structural accuracy and online cost. The dog-bone profile has overall length 165 mm, grip width 19 mm, narrow-section width 13 mm, narrow-section length 57 mm, and fillet radius 76 mm. It is a plane-strain computational benchmark using an ASTM-inspired in-plane profile, not a simulation certified to reproduce an ASTM tensile test.

The two end faces receive equal-and-opposite axial resultants of 100 kN per end, applied in 20 equal increments (Fig. 13). Three point constraints remove rigid modes: the axial and transverse displacement at the left mid-edge node, and transverse displacement at the right mid-edge node. Entire end faces are not clamped. Consequently, a small skew of the deformed ends is not ruled out by the boundary conditions. The anisotropic response can couple axial extension to shear.

The macro mesh contains 690 quadratic triangles, 1501 nodes, and 2070 integration points. The reference extrusion thickness used to convert the two-dimensional stress integrals into the stated force resultants is 50 mm. This force convention is part of the plane-strain benchmark, not an ASTM specimen-thickness specification. The physical cell size is declared as 1.3 mm, corresponding to ten cells across the narrow width. This provides a scale-separation assumption, not a direct numerical validation of homogenization error at that ratio. All saved converged integration-point strains for every reported model remain inside the unchanged training box. Figure 14 shows the reference fields at the final load. The equivalent stress is nearly uniform in the narrow section, peaks slightly where the fillets begin, and decreases toward the grips.

Error and timing definitions. For a saved nodal or integration-point array A , the reported relative error is

$$e_A = 100 \frac{\|\text{vec}(A - A_0)\|_2}{\|\text{vec}(A_0)\|_2}, \quad (68)$$

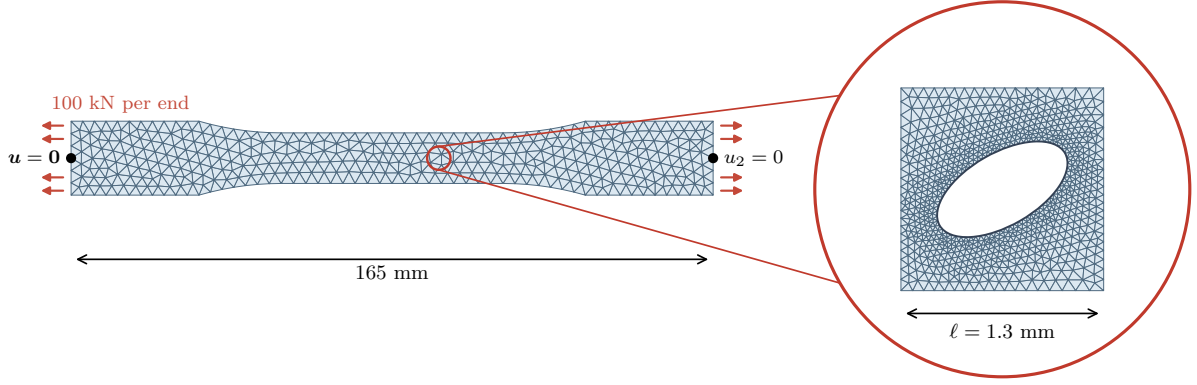


Figure 13: Coupon problem. Undeformed macro mesh with equal-and-opposite resultants of 100 kN per end; the three point constraints only remove rigid-body motion, since the two resultants balance each other. Every integration point queries the SC-RVE, solved by the FOM or replaced by a reduced model or a learned energy.

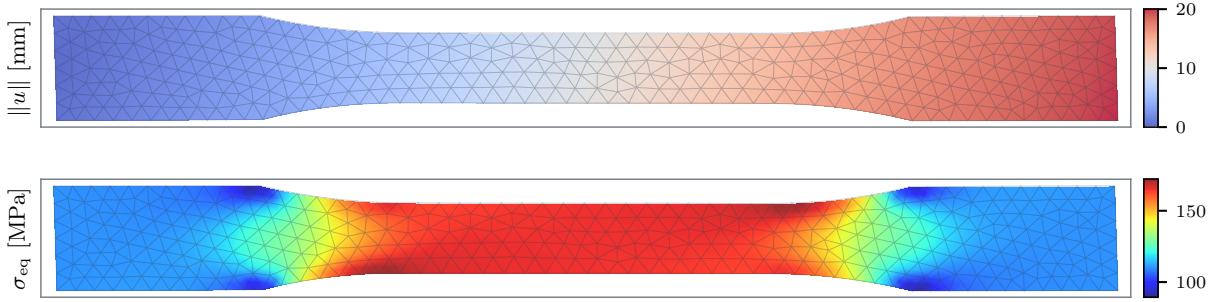


Figure 14: Nested FOM-FE² reference at 100 kN per end, shown on the deformed mesh without displacement amplification. Top: displacement magnitude. Bottom: in-plane von Mises equivalent of the macroscopic Cauchy stress, $\sigma_{eq} = (\sigma_{11}^2 - \sigma_{11}\sigma_{22} + \sigma_{22}^2 + 3\sigma_{12}^2)^{1/2}$; Gauss-point values are averaged to the nodes for visualization.

where A_0 is the nested reference on the same meshes. These are unweighted discrete array norms. The strain array stores engineering shear, whereas the stress array stores tensor shear. They are neither quadrature-weighted continuum L^2 norms nor invariant tensor Frobenius norms. The metric is kept identical across models; this does not make the three different field errors interchangeable.

Online wall time includes the complete macroscopic solve, microscopic or surrogate stress/tangent evaluations, communication, and lazy worker construction within the timed solve. Parent setup, offline data generation, training, reduced-mesh construction outside the timed region, and result export are excluded. The FOM uses 20 worker processes with one numerical-library thread per worker. The three optimized microscopic surrogates use the same process configuration and three sequential fresh complete runs. PANN evaluations are batched in one process, with the number of PyTorch threads selected by a preliminary sweep.

The macroscopic solve stops when the free-degree-of-freedom residual norm falls below 10^{-5} N or its ratio to the initial increment residual falls below 10^{-7} . The periodic FOM uses a relative microscopic displacement-increment tolerance of 10^{-10} , with continuation when needed. The FOM and PANN times come from single runs, so their ratios are observed online speed-ups under the stated implementations rather than matched-thread efficiency estimates.

The affine HPROM uses its exact reduced residual Jacobian, whereas HPROM-ANN uses the projected-stiffness modified-Newton matrix described in Section 4.4.2; both use a microscopic increment tolerance of 10^{-10} . Their effective constitutive tangents nevertheless account for the equilibrium sensitivity of their respective full residuals. The reported speed-up compares these implemented workflows, including their different iteration matrices and derivative costs; it does not isolate a nonlinear-representation gain or quantify the iteration penalty of modified Newton.

Table 5: Coupon simulations: final-field errors of Eq. (68) with respect to the nested FOM, online wall time, and speed-up over the FOM.

Model	e_u [%]	e_E [%]	e_S [%]	Time [s]	Speed-up
FOM	—	—	—	4268.68	—
HPROM	0.0057	0.0165	0.0010	67.35	63.38
HPROM-ANN	0.0520	0.1497	0.0373	30.14	141.61
D-HPROM-ANN	0.0232	0.0485	0.0097	3.79	1126.90
ICNN	0.4066	0.6785	0.0739	1.04	4105.14
ICKAN	0.4197	0.6914	0.0732	1.95	2187.66
Unconstrained energy	0.0022	0.0050	0.0006	1.48	2884.31

Accuracy–cost trade-off and Newton behavior. Table 5 shows distinct operating points rather than a single universally best model. Among the reduced microscopic models, the linear HPROM gives the smallest displacement error, approximately 0.0057%, at a median online time of 67.35 s. The iterative HPROM-ANN reduces this time to 30.14 s with a displacement error of approximately 0.0520%. The iterative nonlinear model is therefore about 2.23 times faster than the linear one, both in their optimized implementations. The direct closure reduces the time to 3.79 s and has a displacement error of approximately 0.0232% on this path. Its lower error than the iterative nonlinear model is an observed outcome, not a consequence of a general error ordering.

The six-feature ICNN and ICKAN take 1.04 s and 1.95 s, with displacement errors of approximately 0.407% and 0.420%, below their constitutive test errors but above those of the reduced microscopic models on this benchmark. The Unconstrained energy takes 1.48 s and gives the smallest displacement error in Table 5, 0.0022%, which mirrors the constitutive ordering of Section 5.3.2. Every saved coupon state lies inside the box, where all three energies have positive rank-one curvature at every sampled state, so this tensile path does not exercise the guarantees that separate them. On this path the constrained energies trade about two orders of magnitude in structural accuracy for admissibility beyond the data, while keeping the displacement error near 0.4%.

All models require four macroscopic iterations per increment, so this example provides no evidence that polyconvexity reduces Newton iterations. Inside the box, the constrained energies thus run in the common structural solver like the Unconstrained energy, at comparable cost. Their guarantees matter at states that the coupon never reaches, and the next example reaches them deliberately.

5.3.6 Confined compression beyond the data

Confined compression reaches such states through compaction. A square block of side 10 mm, meshed with 128 quadratic triangles, is compressed by a uniform traction on one face while rigid, frictionless walls fix the normal displacement of the other three (Fig. 15a). The three SC-RVE energies, the assembly, and the Newton iteration are those of the coupon, and the load is applied in 20 equal increments up to 500 MPa, in x or in y . Compression in x leaves the fitting box with the first increment, since the box contains no axial compression, and compression in y leaves it beyond about 10% shortening. At every converged increment we also evaluate the rank-one curvature of Eq. (69) at the 384 Gauss points of the block, minimized over unit directions. Where it is positive, the incremental equilibrium problem is strongly elliptic.

Both constrained energies reach 500 MPa in both directions, with four to nine Newton iterations per increment and final shortenings of about 22% in x and 31% in y (Fig. 15b,c). The Unconstrained energy follows a softer curve in x and nearly the same curve in y , until Newton does not converge within 50 iterations in the increment to 250 MPa in x and to 225 MPa in y . This stop coincides with a loss of rank-one convexity in the block (Fig. 15d,e). The minimum curvature of the Unconstrained energy falls steadily with the load and turns negative at the last converged increment, at one of the 384 Gauss points in x and at two in y . These points lie in the elements at the corner where the left and bottom walls meet, where the local volume ratio has fallen to J between 0.61 and 0.72. The constrained energies, rank-one convex by construction, keep the minimum curvature above 194 MPa at every Gauss point and increment. The test does not compare accuracy, because neither data nor a reference exist at these states; it shows which energies keep the structural problem solvable once it leaves the data.

6 Conclusions

Paired directional features with constrained trainable powers and determinant exponents provide a practical polyconvex constitutive construction for the anisotropic periodic cells studied here. Their isotropic reference

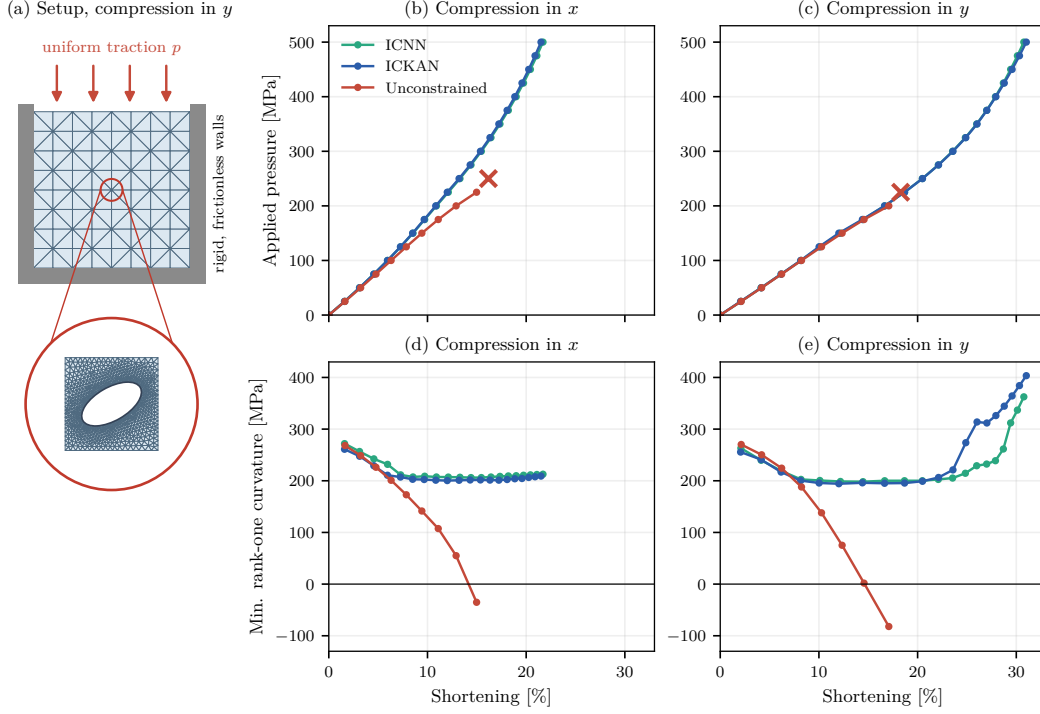


Figure 15: Confined compression beyond the data with the SC-RVE energies. (a) Setup for compression in y : the homogenized block, meshed with 128 quadratic triangles, evaluates the SC-RVE energy at every integration point; for compression in x the traction acts on a vertical face. (b, c) Applied pressure versus shortening, one marker per increment of 25 MPa; a cross marks the increment in which Newton does not converge. (d, e) Minimum rank-one curvature over the 384 Gauss points of the block at each converged increment.

derivatives allow exact stress normalization through an affine determinant correction without tying the normalization coefficient to the volumetric barrier stiffness. The proof combines feature convexity in independent minors with core convexity and monotonicity, and includes every correction in the complete energy. Global nonnegative energy is supported separately by a sufficient bound evaluated for the saved parameters, which certifies every learned six-feature model of this study.

The two cells separate representation from deployment. On the multicavity cell, learning six paired features reduces the held-out stress error sevenfold, to about 0.04%, and prescribing more directions does not close this gap. On the single-cavity cell, the selected ICNN and ICKAN attain approximately 0.63% held-out stress error and below 0.42% displacement error in the coupon simulation. The Unconstrained energy, trained under the same protocol, is more accurate inside the data by a factor of about six on the first cell and about 120 on the second, and controls of capacity and feature count attribute this price to the admissible construction rather than to network size. Beyond the data, the Unconstrained energy loses rank-one convexity and remains finite under volume collapse, whereas the constrained energies are rank-one convex everywhere and grow without bound as the volume vanishes.

Optimized reduced microscopic models provide a complementary accuracy–cost reference. They are more accurate than the constrained energies on the coupon, while direct learned constitutive evaluation is generally faster, and all compared models converge with the same number of macroscopic iterations per increment. Because every coupon state lies inside the data, that example does not exercise the guarantees, and the Unconstrained energy is the most accurate model there. Confined compression takes the same energies beyond the data. The constrained energies then carry the block to 500 MPa in both directions, whereas with the Unconstrained energy Newton stops below half that load, exactly when its rank-one curvature turns negative at a corner of the block. The accuracy given up is thus the price of a structural problem that remains solvable beyond the data, which is the property the construction is designed to secure.

A Rank-one paths and finite-strain energy curvature

This appendix supplies the algebra behind the two finite-strain curvature statements in criterion (C5). For a twice differentiable energy, nonnegative curvature along every rank-one perturbation is expressed by the Legendre–Hadamard condition

$$\partial_{\mathbf{F}\mathbf{F}}^2 W(\mathbf{F})[\mathbf{a} \otimes \mathbf{b}, \mathbf{a} \otimes \mathbf{b}] \geq 0 \quad \text{for all } \mathbf{a}, \mathbf{b} \in \mathbb{R}^2. \quad (69)$$

The argument below shows why the polyconvex representation in Eq. (18) implies this condition.

Determinant along a rank-one path. For arbitrary $\mathbf{F}, \mathbf{H} \in \mathbb{R}^{2 \times 2}$, expansion of the two products in the determinant gives

$$\begin{aligned} \det(\mathbf{F} + t\mathbf{H}) &= (F_{11} + tH_{11})(F_{22} + tH_{22}) - (F_{12} + tH_{12})(F_{21} + tH_{21}) \\ &= \det \mathbf{F} + t(F_{22}H_{11} + F_{11}H_{22} - F_{21}H_{12} - F_{12}H_{21}) + t^2 \det \mathbf{H}. \end{aligned} \quad (70)$$

The coefficient of t is $\text{cof } \mathbf{F} : \mathbf{H}$, since

$$\text{cof } \mathbf{F} = \begin{pmatrix} F_{22} & -F_{21} \\ -F_{12} & F_{11} \end{pmatrix}, \quad \mathbf{A} : \mathbf{B} = \sum_{i,j} A_{ij} B_{ij}. \quad (71)$$

For $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, its entries are $H_{ij} = a_i b_j$, so

$$\det \mathbf{H} = a_1 b_1 a_2 b_2 - a_1 b_2 a_2 b_1 = a_1 a_2 (b_1 b_2 - b_2 b_1) = 0. \quad (72)$$

The determinant is therefore affine along a rank-one path:

$$\det(\mathbf{F} + t\mathbf{a} \otimes \mathbf{b}) = \det \mathbf{F} + t \text{cof } \mathbf{F} : (\mathbf{a} \otimes \mathbf{b}). \quad (73)$$

If $\mathbf{F}_B = \mathbf{F}_A + \mathbf{H}$ and $\mathbf{F}_s = (1-s)\mathbf{F}_A + s\mathbf{F}_B$, with $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, then

$$(\mathbf{F}_s, \det \mathbf{F}_s) = (1-s)(\mathbf{F}_A, \det \mathbf{F}_A) + s(\mathbf{F}_B, \det \mathbf{F}_B). \quad (74)$$

For $0 \leq s \leq 1$ and positive endpoint determinants, the whole segment remains admissible. Convexity of $\mathcal{P}$ therefore yields

$$W(\mathbf{F}_s) \leq (1-s)W(\mathbf{F}_A) + sW(\mathbf{F}_B). \quad (75)$$

For a twice differentiable energy, the restriction to this line consequently has nonnegative second derivative, as required by Eq. (69).

Strain curvature and the chain rule. Let $\mathbf{H} \in \mathbb{R}^{2 \times 2}$ be an arbitrary perturbation and let t parametrize $\mathbf{F}(t) = \mathbf{F} + t\mathbf{H}$. Dots denote derivatives with respect to t , and $\text{sym}(\mathbf{A}) = (\mathbf{A} + \mathbf{A}^T)/2$ denotes the symmetric part of a matrix. Direct multiplication gives

$$\begin{aligned} \mathbf{E}(t) &= \frac{1}{2} [\mathbf{F}^T \mathbf{F} - \mathbf{I} + t(\mathbf{F}^T \mathbf{H} + \mathbf{H}^T \mathbf{F}) + t^2 \mathbf{H}^T \mathbf{H}] \\ &= \mathbf{E}(0) + t \text{sym}(\mathbf{F}^T \mathbf{H}) + \frac{1}{2} t^2 \mathbf{H}^T \mathbf{H}. \end{aligned} \quad (76)$$

Thus $\dot{\mathbf{E}}(0) = \text{sym}(\mathbf{F}^T \mathbf{H})$ and $\ddot{\mathbf{E}}(0) = \mathbf{H}^T \mathbf{H}$. Using the engineering-strain convention of Eq. (2), the chain and product rules give

$$\begin{aligned} \frac{dW}{dt} &= \mathbf{s} \cdot \dot{\mathbf{e}}, \\ \frac{d^2 W}{dt^2} &= \sum_{i,j} \frac{\partial s_i}{\partial e_j} \dot{e}_j \dot{e}_i + \sum_i s_i \ddot{e}_i = \dot{\mathbf{e}}^T \mathbf{D} \dot{\mathbf{e}} + \mathbf{S} : \ddot{\mathbf{E}}. \end{aligned} \quad (77)$$

Substituting $\ddot{\mathbf{E}}(0)$ gives

$$\left. \frac{d^2 W(\mathbf{F} + t\mathbf{H})}{dt^2} \right|_{t=0} = \dot{\mathbf{e}}(0)^T \mathbf{D} \dot{\mathbf{e}}(0) + \mathbf{S} : (\mathbf{H}^T \mathbf{H}). \quad (78)$$

Here $\mathbf{D} = \partial \mathbf{s} / \partial \mathbf{e}$ is the strain tangent and $\mathbf{S}$ is the second Piola–Kirchhoff stress, both evaluated at $t = 0$. The first term accounts for the change of stress with strain; the second accounts for the curvature of the strain path, weighted by the current stress. Even if $\mathbf{D}$ is positive semidefinite, the stress term can be negative in compression.

An analytical counterexample. Consider $W(\mathbf{E}) = \mu \mathbf{E} : \mathbf{E}$, where $\mu > 0$ is a constant with units of energy density. One has $\mathbf{S} = 2\mu \mathbf{E}$. Since

$$W(\mathbf{e}) = \mu(e_1^2 + e_2^2 + e_3^2/2), \quad \mathbf{D} = \text{diag}(2\mu, 2\mu, \mu), \quad (79)$$

the energy is strictly convex in engineering strain. At $\mathbf{F} = \lambda \mathbf{I}$, where $\lambda > 0$ is the common in-plane stretch, choose the rank-one perturbation $\mathbf{H} = \text{diag}(1, 0)$. Then

$$\dot{\mathbf{e}} = (\lambda, 0, 0)^T, \quad \mathbf{S} = \mu(\lambda^2 - 1)\mathbf{I}, \quad \mathbf{H}^T \mathbf{H} = \text{diag}(1, 0). \quad (80)$$

Hence the strain-tangent term is $2\mu\lambda^2$, and the stress term is $\mu(\lambda^2 - 1)$, giving

$$\left. \frac{d^2 W(\lambda \mathbf{I} + t \mathbf{H})}{dt^2} \right|_{t=0} = 2\mu\lambda^2 + \mu(\lambda^2 - 1) = \mu(3\lambda^2 - 1). \quad (81)$$

At $\lambda = 1/2$, the curvature is $-\mu/4$, violating Eq. (69) despite convexity in $\mathbf{E}$. This is an analytical counterexample, not a sampled observation about the RVE or a trained model. Compression alone does not force negative curvature: at $\lambda = 3/4$, the same expression is $11\mu/16 > 0$.

Equivalently, direct substitution gives

$$W(\lambda \mathbf{I} + t \mathbf{H}) = \frac{\mu}{4}[(\lambda + t)^2 - 1]^2 + \frac{\mu}{4}(\lambda^2 - 1)^2, \quad (82)$$

whose second derivative at zero is $\mu(3\lambda^2 - 1)$. The path is admissible near zero because $\det(\lambda \mathbf{I} + t \mathbf{H}) = \lambda(\lambda + t) > 0$ for $t > -\lambda$.

B Proofs for the paired features and complete energy

The proof follows the construction used in the main text. We first isolate one stretch-determinant contribution and derive the exponent bounds that make it convex. We then assemble two perpendicular contributions and compute their reference derivative, proving Proposition 1. Finally, we compose the resulting features with the monotone convex core and examine W_{learn} , W_{corr} , W_{vol} , and W_{dist} to establish Proposition 2.

B.1 Joint convexity of the directional building block

Every contribution in Eq. (25) has the form

$$f(\mathbf{x}, J) = \|\mathbf{x}\|^a J^{-b}, \quad \mathbf{x} = \mathbf{F} \mathbf{d}, \quad (83)$$

up to a positive constant, with $a = 2p$ or $2q$. It is therefore enough to determine when this building block is jointly convex in $(\mathbf{x}, J)$. Let $t = \|\mathbf{x}\| > 0$. Because f depends on $\mathbf{x}$ only through t , its Hessian separates into a block coupling the radial variable t to J and transverse directions orthogonal to $\mathbf{x}$. The radial-determinant block is

$$\begin{pmatrix} a(a-1)t^{a-2}J^{-b} & -abt^{a-1}J^{-b-1} \\ -abt^{a-1}J^{-b-1} & b(b+1)t^a J^{-b-2} \end{pmatrix}. \quad (84)$$

For this symmetric 2×2 block to be positive semidefinite, its diagonal entries and determinant must be nonnegative. The diagonal entries are nonnegative for $a \geq 1$ and $b \geq 0$, while the determinant is

$$\begin{aligned} \det &= ab[(a-1)(b+1) - ab]t^{2a-2}J^{-2b-2} \\ &= ab(a-b-1)t^{2a-2}J^{-2b-2} \geq 0. \end{aligned} \quad (85)$$

The factor $a-b-1$ gives the upper bound $b \leq a-1$; if this bound is violated, the determinant is negative and the block has curvatures of opposite sign. The remaining, transverse Hessian eigenvalues are $at^{a-2}J^{-b} \geq 0$. Hence $a \geq 1$ and $0 \leq b \leq a-1$ make the complete Hessian positive semidefinite. Continuous extension gives convexity at $\mathbf{x} = 0$. For the constitutive evaluation on $\text{GL}^+(2)$, a nonzero reference direction cannot satisfy $\mathbf{F} \mathbf{d} = 0$, so this point does not introduce an interior singularity of the directional norm. Substituting $a = 2p$ gives $p \geq \frac{1}{2}$ and $0 \leq b \leq 2p-1$; the same argument with (q, c) establishes all bounds in Eq. (26).

B.2 Proof of Proposition 1

Proof. Convexity and objectivity. For any fixed learned parameter set, the directions $\mathbf{d}_i$ and $\mathbf{d}_i^\perp$ are independent of deformation. The maps $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i$ and $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i^\perp$ are therefore linear. Composing the convex building block above with these maps, multiplying by the positive factors $1/p_i$ and $1/q_i$, and adding the two contributions proves convexity of z_i in independent $(\mathbf{F}, J)$. A left proper rotation changes $\mathbf{F}\mathbf{d}$ to $\mathbf{Q}\mathbf{F}\mathbf{d}$ without changing its norm, so each contribution is objective.

Reference value and derivative. We now return to the physical relation $J = \det \mathbf{F}$, still holding every learned parameter fixed. At $\mathbf{E} = \mathbf{0}$, both directional invariants and J equal one, so the two contributions give $z_i^0 = 1/p_i + 1/q_i$. Moreover, $I_4(\mathbf{d}_i) = \mathbf{d}_i^T(\mathbf{I} + 2\mathbf{E})\mathbf{d}_i$ and $J^2 = (1 + 2E_{11})(1 + 2E_{22}) - 4E_{12}^2$ imply $\partial_{\mathbf{E}} I_4(\mathbf{d}_i) = 2\mathbf{d}_i \otimes \mathbf{d}_i$ and $\partial_{\mathbf{E}} J|_0 = \mathbf{I}$. Applying the product rule to the first contribution gives

$$\partial_{\mathbf{E}} \frac{I_4(\mathbf{d}_i)^{p_i}}{p_i J^{b_i}} \Big|_0 = \frac{1}{p_i} [2p_i \mathbf{d}_i \otimes \mathbf{d}_i - b_i \mathbf{I}] = 2\mathbf{d}_i \otimes \mathbf{d}_i - \frac{b_i}{p_i} \mathbf{I}. \quad (86)$$

The prefactor $1/p_i$ has cancelled the factor p_i produced by differentiating the stretch power. The perpendicular contribution has the same form with $(\mathbf{d}_i, p_i, b_i)$ replaced by $(\mathbf{d}_i^\perp, q_i, c_i)$. The two directional projectors satisfy $\mathbf{d}_i \otimes \mathbf{d}_i + \mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp = \mathbf{I}$; adding the contributions therefore removes the directional dependence at the reference and gives

$$\partial_{\mathbf{E}} z_i|_0 = 2(\mathbf{d}_i \otimes \mathbf{d}_i + \mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp) - \left(\frac{b_i}{p_i} + \frac{c_i}{q_i} \right) \mathbf{I} = \rho_i \mathbf{I}. \quad (87)$$

In the work-conjugate engineering-strain convention of Eq. (2), this tensor gradient corresponds to $\partial_{\mathbf{e}} z_i|_0 = \rho_i(1, 1, 0)^T$, proving Eq. (28). $\square$

Positive affine conditioning. The raw feature is converted to the core input by $y_i = (z_i - z_i^{\text{cen}})/s_i$, with $s_i > 0$ independent of deformation. Writing $\mathbf{X} = (\mathbf{F}, J)$,

$$\partial_{\mathbf{X}\mathbf{X}}^2 y_i = \frac{1}{s_i} \partial_{\mathbf{X}\mathbf{X}}^2 z_i, \quad y_i(\mathbf{X}_B) - y_i(\mathbf{X}_A) = \frac{z_i(\mathbf{X}_B) - z_i(\mathbf{X}_A)}{s_i}. \quad (88)$$

The first identity preserves positive-semidefinite curvature and the second preserves componentwise order. Thus the numerical conditioning applied before the monotone core changes neither property used in the composition proof below.

B.3 Proof of Proposition 2

Proof. Potential structure, objectivity, and stress symmetry (C1)–(C3). Every directional norm and J is unchanged by left proper rotations. The features can also be expressed through $\mathbf{d}_i^T \mathbf{C} \mathbf{d}_i$ and $\sqrt{\det \mathbf{C}}$. Hence the complete energy is objective, and differentiation with respect to symmetric strain gives a symmetric material stress. Differentiating this scalar energy supplies (C1) and the major symmetry of its strain tangent. Smoothness holds on $\text{GL}^+(2)$ because $J > 0$, nonzero material directions have nonzero deformed length, and both cores are C^2 . Because this argument uses left rotations only, it does not impose a material symmetry group.

Reference normalization (C4). Equation (38) shows that W_{learn} sets the reference energy, and Eq. (40) shows that W_{corr} cancels the learned reference stress. The calculations for W_{vol} and W_{dist} in Section 3.5 show that both growth terms have zero value and zero strain gradient at the reference. Thus the complete energy satisfies both reference conditions.

Polyconvexity (C5). Let $\mathbf{G} \in \mathbb{R}^{2 \times 2}$ and $j > 0$ be independent. Define $\mathcal{P}(\mathbf{G}, j)$ by the right-hand side of Eq. (37), replacing $\mathbf{z}$ by $\mathbf{z}(\mathbf{G}, j)$, J by j , and $\text{tr } \mathbf{C}$ by $\|\mathbf{G}\|_F^2$. For two such pairs $\mathbf{X}_1 = (\mathbf{G}_1, j_1)$, $\mathbf{X}_2 = (\mathbf{G}_2, j_2)$ and $0 \leq t \leq 1$,

$$\begin{aligned} H(\mathbf{z}(t\mathbf{X}_1 + (1-t)\mathbf{X}_2)) &\leq H(t\mathbf{z}(\mathbf{X}_1) + (1-t)\mathbf{z}(\mathbf{X}_2)) \\ &\leq tH(\mathbf{z}(\mathbf{X}_1)) + (1-t)H(\mathbf{z}(\mathbf{X}_2)). \end{aligned} \quad (89)$$

The first inequality uses feature convexity and core monotonicity; the second uses core convexity. Therefore W_{learn} is convex up to its constant reference subtraction. The correction $W_{\text{corr}} = -r(j-1)$ is affine for either sign of r , while W_{vol} and W_{dist} are convex in independent $(\mathbf{G}, j)$ because their nonaffine parts are positive multiples of $-\log j$, $(j-1)^2$, and $\|\mathbf{G}\|_F^2$. Thus $\mathcal{P}$ is convex and $W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F})$, establishing Eq. (18).

Growth (C6). Convexity of H gives its supporting-plane inequality at $\mathbf{z}^0$:

$$H(\mathbf{z}) \geq H(\mathbf{z}^0) + \sum_i h_i(z_i - z_i^0) \geq H(\mathbf{z}^0) - \sum_i h_i z_i^0. \quad (90)$$

The second inequality uses $h_i, z_i \geq 0$. Hence W_{learn} has a deformation-independent lower bound and cannot cancel the analytic growth terms. As $J \rightarrow 0^+$, the logarithmic part of $W_{\text{vol}} + W_{\text{dist}}$ diverges while W_{corr} remains bounded. As $J \rightarrow \infty$, the positive quadratic part of W_{vol} dominates every potentially negative linear or logarithmic term. Finally, if J remains in a compact subset of $(0, \infty)$ while $\|\mathbf{F}\|_F \rightarrow \infty$, the term $\varepsilon \|\mathbf{F}\|_F^2/2$ in W_{dist} diverges and all determinant-only terms remain bounded. These are the three stated growth limits. $\square$

C Sufficient lower bound for nonnegative energy

Define $k_i = 1/p_i + 1/q_i$ and $\eta_i = \rho_i/k_i$. Write $t = \|\mathbf{F}\mathbf{d}_i\|^2$, $u = \|\mathbf{F}\mathbf{d}_i^\perp\|^2$. Hadamard's determinant inequality gives $tu \geq J^2$. Since each feature is increasing in t, u , its minimum at fixed J occurs on $tu = J^2$. Minimizing

$$\frac{t^{p_i}}{p_i J^{b_i}} + \frac{J^{2q_i - c_i} t^{-q_i}}{q_i} \quad (91)$$

gives $t^{p_i + q_i} = J^{2q_i + b_i - c_i}$ and hence $z_i \geq k_i J^{\eta_i}$. This is a separate lower bound for each feature; simultaneous attainability for all directions is not assumed. It therefore remains valid after summing the supporting-plane inequalities.

Together with $\text{tr } \mathbf{C}/2 \geq J$ and the supporting plane of H , this feature bound gives

$$W \geq g(J) := \sum_i h_i k_i [J^{\eta_i} - 1 - \eta_i(J - 1)] + (\alpha + \varepsilon)(J - 1 - \log J) + \frac{\beta}{2}(J - 1)^2. \quad (92)$$

Because $g(1) = g'(1) = 0$, it remains to establish $g''(J) \geq 0$ for every $J > 0$. Differentiation gives

$$J^2 g''(J) = \alpha + \varepsilon + \beta J^2 + \sum_i c_i^* J^{\eta_i}, \quad c_i^* = h_i k_i \eta_i (\eta_i - 1). \quad (93)$$

Only terms with $0 < \eta_i < 1$ can have negative coefficients. If $c_- = \sum_{c_i^* < 0} |c_i^*|$, the two inequalities

$$\alpha + \varepsilon + \sum_{\eta_i < 0} c_i^* - c_- \geq 0, \quad (94)$$

$$\beta + \sum_{\eta_i \geq 1} c_i^* - c_- \geq 0 \quad (95)$$

are sufficient on $0 < J \leq 1$ and $J \geq 1$, respectively. For the first region, $J^{\eta_i} \leq 1$ for negative-coefficient terms and $J^{\eta_i} \geq 1$ when $\eta_i < 0$. For the second region, divide $J^2 g''$ by J : all negative-coefficient powers then satisfy $J^{\eta_i - 1} \leq 1$, while $J^{\eta_i - 1} \geq 1$ for $\eta_i \geq 1$ and $\beta J \geq \beta$. Dropping the remaining nonnegative contributions gives the stated bounds. Their failure is inconclusive.

The selected parameters require a sharper check. The audit partitions $[10^{-4}, 10^4]$ into 4000 logarithmically spaced intervals. On each interval $[J_L, J_R]$, a positive power term is bounded below by the smaller endpoint value, whereas a negative-coefficient term is bounded below by $c_i^* J_R^{\eta_i}$, since its exponent lies in $(0, 1)$. The constant and quadratic terms are bounded below by $\alpha + \varepsilon$ and βJ_L^2 . Below the finite interval, the nonnegative negative-exponent terms increase and the magnitudes of the negative-coefficient terms decrease toward $J = 0$. Above it, factoring out J^{η_-} with $\eta_- = \max_{c_i^* < 0} \eta_i < 1$ yields a lower bound whose retained positive powers increase. These endpoint and tail inequalities cover every $J > 0$, not only the grid points. Positive margins establish the sufficient analytical argument for the saved parameters to the accuracy of its floating-point evaluation; random energy samples alone would not establish it.

D Latent-space closure details

This appendix supplies the algebra omitted from Section 4.3. Let $\Phi \in \mathbb{R}^{N \times n_{\text{tot}}}$ be the retained orthonormal POD basis, let $\mathbf{S}_{\text{snap}}$ collect the centered displacement snapshots, and collect the corresponding engineering-strain inputs in $\mathbf{M} = [\mathbf{e}^1, \dots, \mathbf{e}^{N_s}]$. Their POD coefficients are

$$\mathbf{Q}_{\text{POD}} = \Phi^T \mathbf{S}_{\text{snap}}. \quad (96)$$

The input-informed construction first identifies the least-squares map from these coefficients to strain,

$$\mathbf{T}_m = \mathbf{M} \mathbf{Q}_{\text{POD}}^\dagger \in \arg \min_H \|\mathbf{H} \mathbf{Q}_{\text{POD}} - \mathbf{M}\|_F^2, \quad (97)$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. Assuming rank three, the compact factorization

$$\Phi \mathbf{T}_m^T = \mathbf{V} \Sigma_m \mathbf{Z}_m^T, \quad \mathbf{A}_m = \Sigma_m^{-1} \mathbf{Z}_m^T \quad (98)$$

defines the three-column orthonormal primary block $\mathbf{V}$ and its coordinate scaling $\mathbf{A}_m$. The secondary block $\bar{\mathbf{V}}$ is an orthonormal complement within the retained POD span, ordered by snapshot energy. With $\mathbf{q} = \mathbf{A}_m \boldsymbol{\xi}$, the deployed decoder is

$$\tilde{\mathbf{u}}(\boldsymbol{\xi}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \boldsymbol{\xi} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\boldsymbol{\xi}), \quad \boldsymbol{\xi}^s = \mathbf{T}_m \mathbf{Q}_{\text{POD}}^s. \quad (99)$$

Thus $\boldsymbol{\xi}^s \approx \mathbf{e}^s$ is a fitted snapshot relation, not a kinematic identity.

The displacement error separates the two approximation mechanisms. For $\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}]$ and projected coordinates $(\mathbf{q}^s, \bar{\mathbf{q}}^s)$,

$$\|\mathbf{u}^s - \tilde{\mathbf{u}}(\mathbf{q}^s)\|_2^2 = \|\bar{\mathbf{q}}^s - \mathcal{N}(\mathbf{q}^s)\|_2^2 + \|(\mathbf{I} - \mathbf{V}_{\text{tot}} \mathbf{V}_{\text{tot}}^T)(\mathbf{u}^s - \mathbf{u}_{\text{ref}})\|_2^2. \quad (100)$$

The first term is the closure error within the retained span; the second is the POD truncation error. Increasing network capacity cannot recover a displacement component excluded from $\mathbf{V}_{\text{tot}}$.

Finally, let $\mathbf{B}_\xi = \partial_{\boldsymbol{\xi}} \tilde{\mathbf{u}}$, $\mathbf{R}$ be the constrained microscopic residual, and $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{R}$. Differentiating the projected residual $\mathbf{r} = \mathbf{B}_\xi^T \mathbf{R}$ gives

$$\frac{\partial \mathbf{r}}{\partial \boldsymbol{\xi}} = \mathbf{B}_\xi^T \mathbf{K} \mathbf{B}_\xi + \sum_{j=1}^N R_j \frac{\partial^2 \tilde{u}_j}{\partial \boldsymbol{\xi} \partial \boldsymbol{\xi}}. \quad (101)$$

The second term is the curvature contribution of the decoder manifold. It need not vanish at reduced equilibrium because $\mathbf{B}_\xi^T \mathbf{R} = \mathbf{0}$ does not imply $\mathbf{R} = \mathbf{0}$.

E Elementwise cubature conventions

This appendix records the implementation conventions omitted from Section 4. The general ECM construction and MAW–ECM pruning algorithm follow Refs. [34, 39]; the expressions below specify how the deployed residual and stress rules are assembled.

Fixed rule. Let $\mathbf{c}_e^s \in \mathbb{R}^d$ contain the contribution of element e at training state s . Residual cubature uses $\mathbf{c}_e^s = \mathbf{h}_e^s$. Stress cubature uses the four components of

$$\mathbf{p}_e^s = \frac{1}{|Y|} \sum_{g \in e} \omega_g \mathbf{P}_{\mu, g}^s, \quad \mathbf{c}_e^s = \text{vec } \mathbf{p}_e^s = [p_{e,11}^s \quad p_{e,12}^s \quad p_{e,21}^s \quad p_{e,22}^s]^T. \quad (102)$$

One element occupies each row of the sampled integrand matrix,

$$\mathbf{M}_{\text{int}} = \begin{bmatrix} (\mathbf{c}_1^1)^T & \cdots & (\mathbf{c}_1^{N_c})^T \\ \vdots & & \vdots \\ (\mathbf{c}_{N_{\text{el}}}^1)^T & \cdots & (\mathbf{c}_{N_{\text{el}}}^{N_c})^T \end{bmatrix} \in \mathbb{R}^{N_{\text{el}} \times (d N_c)}. \quad (103)$$

A truncated SVD gives $\mathbf{M}_{\text{int}} \simeq \mathbf{U}_{\text{cub}} \Sigma_{\text{cub}} \mathbf{V}_{\text{cub}}^T$. To enforce the multiplier sum without duplicating a constant component already represented by the retained integration modes, the fitting system uses

$$\mathbf{p}_\perp = (\mathbf{I} - \mathbf{U}_{\text{cub}} \mathbf{U}_{\text{cub}}^T) \mathbf{1}, \quad \mathbf{G}_{\text{cub}} = \begin{bmatrix} \mathbf{U}_{\text{cub}}^T \\ \mathbf{p}_\perp^T / \|\mathbf{p}_\perp\|_2 \end{bmatrix}, \quad \mathbf{b}_{\text{cub}} = \mathbf{G}_{\text{cub}} \mathbf{1}. \quad (104)$$

If the constant already lies in the retained span, the last row is redundant. Otherwise its exact reproduction gives $\sum_e w_e = N_{\text{el}}$. The reference quadrature measures are already contained in $\mathbf{c}_e^s$; the unknowns are element multipliers.

Adaptive neural fields. After MAW–ECM pruning selects a support $\mathcal{Z} = \{e_1, \dots, e_m\}$, neural fitting returns to the uncompressed contributions at each decoded state:

$$\mathbf{A}^s = \begin{bmatrix} \mathbf{c}_{e_1}^s & \cdots & \mathbf{c}_{e_m}^s \\ 1 & \cdots & 1 \end{bmatrix}, \quad \mathbf{b}^s = \begin{bmatrix} \sum_{e \in \mathcal{Z}} \mathbf{c}_e^s \\ N_{\text{el}} \end{bmatrix}. \quad (105)$$

Thus the target uses the full-mesh sum, not the selected-column sum. Residual fitting uses $d = n$ and $\mathcal{Z}_r$; stress fitting uses $d = 4$ and $\mathcal{Z}_s$. With the scaled-softmax map of Eq. (59), the residual network minimizes

$$\mathcal{L}_{\text{cub}}(\boldsymbol{\theta}_r) = \frac{1}{|\mathcal{I}_{\text{fit}}|} \sum_{s \in \mathcal{I}_{\text{fit}}} \frac{\|\mathbf{A}^s \mathbf{w}(\boldsymbol{\xi}^s) - \mathbf{b}^s\|_2^2}{\max(\|\mathbf{b}^s\|_2^2, \delta_{\text{floor}})}, \quad (106)$$

and the stress network uses the analogous loss with $\mathbf{v}$. A preliminary fit to the sampled MAW–ECM weights initializes each network, but the final objective is integration accuracy; a reproducing weight vector is not unique.

Tangent evaluation. Both reduced implementations use Eq. (62), but evaluate its partial derivatives differently. For the affine HPROM, the assembled reduced stiffness and the analytic lifting derivative provide $d\mathbf{q}_{\text{tra}}^*/d\mathbf{e}$ through three linear right-hand sides; central differences are applied only to the stress along those linearized equilibrium directions. For HPROM–ANN, central differences at fixed $\mathbf{q}$ or fixed $\mathbf{e}$ evaluate all four partial-derivative blocks in Eq. (62), including decoder curvature and weight variation. The resulting small linear system does not require a new nonlinear microscopic solve for every perturbed strain. Numerical differentiation is an implementation choice; analytic or automatic differentiation would give the same formal tangent subject to their respective numerical tolerances.

F Capacity and feature-count controls

The comparisons of Section 5 pair compact constrained cores with a larger Unconstrained network. To separate the effect of the constraints from that of capacity, both benchmarks complete a two-by-two design. The constrained cores are widened to about 25 000 parameters, comparable with the Unconstrained energy, and an Unconstrained energy with 981 parameters matches the compact ICNN. Every new fit follows the protocol of Section 5.1.1 with the same three seeds, and its test states are opened only once, after training has ended.

Table 6 reports the MC-RVE cells as medians over three seeds, as in Table 3. Widening changes the ICNN error from 0.0445% to 0.0398%, while the Unconstrained energy improves from 0.0126% to 0.0073%. At matched capacity the constraints therefore cost a factor of 3.2 to 5.5 in stress error. The wide ICKAN is less accurate than the compact one because two of its three seeds met the validation-plateau criterion early, so its row reflects the common stopping rule rather than capacity.

Table 7 reports the SC-RVE cells for the validation-selected seed, as in Table 4, together with the coupon of Section 5.3.5. Widening changes the constrained test errors by less than 3%, and the Unconstrained energy with 981 parameters remains about 75 to 80 times more accurate than either wide core. In the coupon, the corresponding factor between matched cells lies between about 115 and 200. Thirty-two learned features, trained with the same protocol, lower the test errors to 0.537% and 0.618% and the coupon errors to 0.343% and 0.347%. They also raise the probe errors to 35.6% and 16.9% (Section 5.3.2), and the sufficient bound of Appendix C fails for both saved parameter sets. The six-feature models therefore remain the deployed choice.

Table 6: MC-RVE capacity controls: independent-test errors, in percent.

Model	Parameters	Stress [%]	Energy [%]	Tangent [%]
ICNN, $m = 6$	975	0.0445	0.0079	0.3851
ICNN, wide	26 919	0.0398	0.0069	0.2962
ICKAN, $m = 6$	1 659	0.0406	0.0066	0.3622
ICKAN, wide*	24 847	0.0744	0.0104	1.8311
Unconstrained, small	981	0.0126	0.0019	0.1579
Unconstrained	25 473	0.0073	0.0011	0.0857

Relative error norm over the 512 test states, median over three seeds. *Adam stopped at the validation plateau after 28 540 and 33 330 steps for two of the three seeds, against 114 910 for the third.

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Table 7: SC-RVE capacity and feature-count controls, in percent.

Model	Parameters	e_S^{test} [%]	e_u [%]
ICNN, $m = 6$	975	0.6214	0.4066
ICNN, wide	26 919	0.6066	0.4162
ICNN, $m = 32$	2 379	0.5370	0.3433
ICKAN, $m = 6$	1 659	0.6359	0.4197
ICKAN, wide	24 847	0.6540	0.4322
ICKAN, $m = 32$	3 999	0.6177	0.3469
Unconstrained, small	981	0.0081	0.0035
Unconstrained	25 473	0.0052	0.0022

Validation-selected seed of each cell. Test: 400 states; e_u : final nodal displacement error of the coupon of Section 5.3.5.

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